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State-of-the-Art in Responsible, Explainable, and Fair AI for Medical Image Analysis

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ABSTRACT Integrating responsible, explainable, and fair artificial intelligence (REF-AI) into medical image analysis has gained significant attention in recent years. This has been driven by the pressing need for ethical, trustworthy, and transparent implementation of AI systems in healthcare. The following review provides a concise overview of REF-AI in the context of medical image analysis. It begins with the fundamental concepts of AI responsibility, explainability, and fairness, followed by a comprehensive taxonomy of over 35 key algorithms and strategies. In addition, it compares methodologies, strengths, and limitations, such as the alignment of AI models with medical standards and the development of interpretable and actionable results for clinicians. Finally, it highlights current trends and proposes directions for future research to further advance the responsible, explainable, and fair application of AI in medical imaging.

INDEX TERMS Artificial Intelligence (AI), Responsible AI, Explainable AI, Fair AI, AI Evaluation Metrics, Medical Imaging

I. INTRODUCTION

Integrating artificial intelligence (AI) into healthcare, particularly in medical image analysis, has made transformative advances in diagnosis, prognosis, treatment planning, and patient care. As these advanced technologies continue to evolve, there is a pressing need for AI systems that are effective, ethical, transparent, trustworthy, and equitable. Responsible, explainable, and fair AI (REF-AI) has emerged as a critical focus in ensuring that AI applications in medical imaging meet these standards. REF-AI emphasizes the importance of aligning AI models with well-established medical standards, making their outputs interpretable and actionable for healthcare professionals, and ensuring fairness in their application across diverse patient populations and healthcare settings. This review explores the foundational principles of REF-AI in medical image analysis, providing a comprehensive overview of the key algorithms and methodologies that drive this field. This review aims to advance the responsible, ex-

plainable, and equitable application of AI in medical imaging by critically assessing the strengths, limitations, and challenges of existing approaches while proposing directions for future research. We will begin by defining REF-AI and exploring its empirical foundations.

A. RESPONSIBLE AI IN MEDICAL IMAGE ANALYSIS

Responsible AI refers to building, implementing, and utilizing AI systems in an ethical, sustainable, and trustworthy manner. This focuses on respecting human values, promoting fairness, and ensuring transparency and accountability while considering societal impacts [1]–[4]. Responsible AI in healthcare encompasses the broader ethical and social responsibilities of developing and deploying AI-powered systems across the healthcare ecosystem. This empowers individuals to comprehend, control, and take responsibility for AI-driven mechanisms [1], [5], [6]. Responsible AI in medical imaging ensures that AI tools are implemented and

used in ways that align with medical ethics, fairness and equity, regulations, patient privacy, and safety standards [7]–[10]. This includes comprehensive and even rigorous testing, validation, and continuous monitoring of AI systems to guarantee that they perform precisely, logically, and safely in real-world clinical settings.

B. EXPLAINABLE AI IN MEDICAL IMAGE ANALYSIS

AI explainability in healthcare refers to AI models' capability to provide clear and understandable explanations of their decisions and predictions to their end-users, such as clinicians, surgeons, and patients [11]–[14]. In AI-powered image analysis [15]–[18], and particularly for medical image analysis, explainable AI is critical because healthcare professionals must trust and understand the AI-powered imaging toolsets before they utilize them for patient care and clinical practices. This could bridge the gap between complex AI models, domain experts, clinicians, patients, and decision-makers by providing insights into how and why a particular diagnosis or prediction was made using the AI models [19]–[23].

C. FAIR AI IN MEDICAL IMAGE ANALYSIS

AI fairness involves addressing biases in data, data acquisition (e.g., imaging machinery), algorithms, and outcomes to prevent discrimination [24]–[26]. In medical image analysis, this involves training and validating AI models on diverse datasets that reflect various demographics, such as sex, race, and ethnicity, alongside social determinants of health (SDOH), such as income and social protection, education, availability of healthcare services, health insurance, and quality of care [27], [28]. This method helps reduce the risk of biased predictions that might unequally impact specific patient groups. Fairness in medical imaging is essential to verify that AI tools provide equal benefits to all patients, regardless of their race, sex, age, economic status, or other factors [29]–[32].

These three principles of responsibility, explainability, and fairness confirm that AI systems in medical imaging are technically and computationally effective, ethically sound, and well-equipped with a list of qualitative and quantitative attributes. These attributes include accountability, lawfulness, traceability, reliability, equity, governability, scalability, availability, explainability, truthfulness, privacy-preserving, and safety (Figure 1). These are essential to gain acceptance among healthcare professionals and patients and seamlessly integrate AI into various clinical workflows.

D. EMPIRICAL FOUNDATIONS

The empirical foundation of responsible, explainable, and fair AI (REF-AI) in medical image analysis lies in creating AI models and mechanisms that are reliable, understandable, and fair in various healthcare settings and patient populations. With that, responsible AI emphasizes safety and ethical alignment by evaluating AI tools against real-world clinical standards to ensure accuracy and reliability. Regular

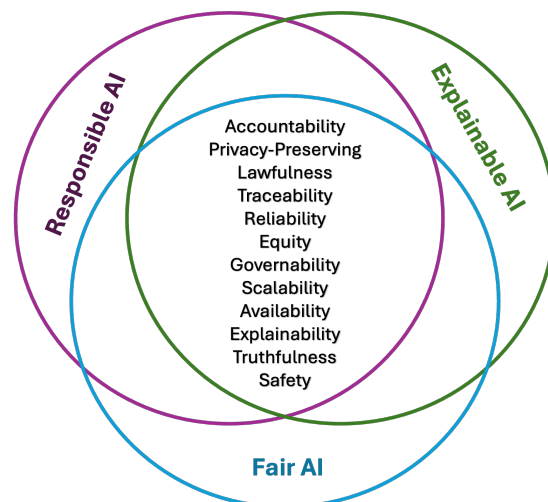


FIGURE 1: Principles for REF-AI in medical imaging include accountability, privacy-preserving, lawfulness, traceability, reliability, equity, governability, scalability, availability, explainability, truthfulness, and safety.

monitoring confirms that AI outputs remain dependable as new patient data or clinical scenarios emerge. Meanwhile, explainable AI makes AI decisions understandable and accessible to healthcare providers and patients, enabling them to comprehend how specific features led to a particular result. This transparency confidently supports clinicians in integrating AI insights into their decision-making. Fair AI, on the other hand, ensures that these systems perform equitably for all patients. Fairness reduces the risk of bias by training AI on diverse datasets representing different ages, ethnicities, and health conditions. It brings consistent performance in all groups, promoting equitable patient care.

Specific metrics and measurements should be used to evaluate each component to assess the effectiveness of REF AI in medical imaging. For AI responsibility, metrics, such as accuracy, sensitivity, and specificity determine if an AI model delivers reliable, accurate, and clinically relevant output. These metrics ensure that AI predictions align with accepted medical standards and provide consistent support in diverse clinical scenarios. Regarding AI explainability, some computational strategies, such as saliency maps [33], [34] and Grad-CAM [35], [36], help clinicians visualize the AI's decision-making process by highlighting the specific areas of a medical image that contributed to a diagnosis. It offers better transparency and allows healthcare providers to interpret AI recommendations relatively easily and quickly. Evaluation tools, such as the Pointing Game [37], [38] or Area Over Perturbation Curve (AOPC) [39] measure how well these visual explanations align with medical insights. The fairness of AI could also be evaluated using fairness-adjusted metrics. For example, using skew equalized odds metric [40], we can compare AI performance across different demographic groups to ensure equitable treatment recommendations. Additionally, skew error ratio (SER) [29] can

measure the discrepancy in error rates between different demographic groups, such as age, gender, or ethnicity, helping to identify potential biases in the AI model's performance. Moreover, metrics like demographic parity and subgroup accuracy help identify potential bias, ensuring that the AI models serve all patient populations effectively and without discrimination.

E. REF-AI APPLICATIONS IN MEDICAL IMAGING

By making AI technologies accurate, efficient, and in line with clinical requirements and regulatory frameworks, REF-AI is now transforming medical imaging into more reliable, more intelligent, and more responsible pipelines in various clinical domains. With reliable predictions, transparent reasoning, and equitable performance, REF-AI provides a basis for safe and practical applications in diagnosis, prognosis, personalized treatment, and imaging, fundamentally enhancing patient care and clinical outcomes.

This section organizes REF-AI applications for medical imaging into five categories, as follows:

- **Diagnosis, Prognosis, and Treatment:** REF-AI supports accurate and equitable diagnostic tools, particularly identifying and predicting disease progression. For instance, REF-AI in diagnostic imaging provides clinicians with clarity by visually highlighting regions of interest (RoI) in medical images, such as cancerous lung nodules, lesions, or abnormal tissue, allowing for faster and more informed clinical decisions. REF-AI techniques improve prognostic predictions while estimating disease progression with high accuracy, which aligns with real-world medical standards. It also makes it possible for clinicians and surgeons to choose treatment paths that reflect patient-specific needs, reducing biases that could adversely impact diagnosis and prognosis across diverse patient demographics [19], [32], [41]–[46].
- **Personalized Medicine:** Personalized care is becoming a reality with REF-AI, as AI systems harness imaging data combined with other patient characteristics, including genetics and medical history, to custom treatment strategies. For example, REF-AI methods clarify the reasoning behind these suggestions, enabling clinicians and patients to understand the factors driving personalized recommendations. These AI-enabled systems make individualized care models available and equally beneficial across all demographic groups, optimizing patient outcomes without compromising equity in care [45]–[51].
- **AI Assistive Radiology:** REF-AI significantly improves radiology by offering AI-based assistance that radiologists can trust and use intuitively in everyday practice. These technologically advanced tools highlight critical areas within radiological images, such as suspicious masses, and bone or prosthesis loosening, enhancing the radiologist's ability to make fast and accurate diagnoses. These AI models are validated against di-

verse clinical imaging datasets to produce consistent and high-quality support, while they are equally accurate and applicable across various patient populations [43], [51]–[58].

- **Medical Image Segmentation and Classification:** REF-AI has made a significant leap towards more actionable and interpretable outcomes in segmentation and classification. REF-AI tools are now helping radiologists and other clinicians understand how AI models segment, measure, and classify regions, such as highlighting tumor boundaries in cancer imaging, which aids in diagnosis and treatment planning. These computational methods focus on reliable and real-world testing, aligning the automatic segmentation algorithms closely with clinical benchmarks and patient-specific characteristics. By reducing biases and making generalized AI models, REF-AI advances image segmentation and classification accuracy across diverse populations, eliminating variability in AI model performance that could otherwise lead to mis-classification or even inconsistent patient outcomes [29], [43]–[45], [52], [59], [60], [60]–[69].
- **Clinical Trials:** REF-AI plays an important in clinical research by supporting unbiased patient selection and reliable outcome prediction, both essential for generalizable and reproducible results. For instance, REF-AI mechanisms assess patient data in a way that reduces demographic or socioeconomic biases, making trials more inclusive and representative. At the same time, these AI-powered methods offer transparency into AI-driven insights built from imaging data, helping researchers interpret these insights and strengthen the credibility of trial outcomes. This application reduces trial dropout rates and enables more precise tracking of intervention effects, advancing research quality and paving the way for broader application in clinical practice [24], [25], [70]–[75].

II. REF-AI CHARACTERISTICS

This section further explains the underlying components of REF-AI in medical imaging. This section will offer a comprehensive explanation of the REF-AI principles depicted in Figure 1.

A. ACCOUNTABILITY

Accountability in REF-AI refers to AI systems that can be audited to meet specific standards, with clearly defined responsibilities and consequences if the systems fail to comply. It also discusses that AI developers, organizations, and policymakers are legally and ethically responsible for the decisions made by AI systems, requiring them to follow laws and standards to ensure AI-powered systems function correctly [76]–[78]. The establishment of REF-AI systems should hold developers and organizations accountable for errors or issues, with accountability mandated at every level from requirement analysis and development to implemen-

tation and deployment [79]. There is also a pressing need to organize a solid data governance framework with regular data audits to assess data records' quantity, quality, and suitability, equipping such a framework with data provenance, data traceability, and a comprehensive data dictionary and documentation [80]. One way to achieve accountability in AI systems for medical imaging is by ensuring AI decisions are transparent and explainable to end-users (e.g., physicians, radiologists, and clinicians). This allows them to review, investigate, and validate the AI's conclusions before conveying information to patients. This collaborative process adds an important layer of human oversight, confirming that AI tool sets are used responsibly, and they can deliver accurate and trustworthy medical guidance [81], [82].

B. PRIVACY-PRESERVING

Privacy-preserving AI in REF-AI refers to developing and deploying AI systems that safeguard sensitive patient information while maintaining the system's ability to deliver reliable, trustworthy, accurate, and actionable insights. Given the highly confidential nature of healthcare data, including medical images combined with clinical notes or radiology reports, ensuring privacy preservation is essential for compliance with legal and ethical standards, such as the Health Insurance Portability and Accountability Act (HIPAA) and General Data Protection Regulation (GDPR). Beyond regulatory compliance, privacy-preserving AI builds trust among stakeholders, including patients, surgeons, clinicians, and healthcare organizations, sharing data with AI algorithms in a way that cannot be traced back to individuals. This trust is fundamental to the successful and widespread integration of AI technologies in medical imaging informatics and healthcare [83]–[85]. Practical methods of privacy-preserving AI are rapidly growing in medical imaging. For example, federated learning and homomorphic encryption have been utilized for training AI models across multiple institutions, allowing collaboration while securing patient data [86]–[90].

The federated learning approach allows AI models to be trained on distributed data sources without transferring data via a central server. This allows sensitive patient data to remain within the institution that owns it [86], [87]. Additionally, with computational methods such as encrypted data processing using homomorphic encryption, we can make computations directly on encrypted data, safeguarding privacy throughout the AI pipeline [88], [89]. Research using AI and associated data bases will exponentially increase; safeguards regarding compliance with national and international rules and regulations regarding human research need to be maintained.

C. LAWFULNESS

Lawfulness in AI refers to aligning AI systems with established legal and regulatory frameworks, guaranteeing adherence to norms and standards that protect patients and uphold ethical principles. In medical imaging, where AI

systems analyze sensitive patient data and assist in clinical decision-making, lawfulness becomes paramount. Compliance with regulations, such as the GDPR in the European Union [91], emphasizes that AI must manage health data records responsibly. The GDPR sets comprehensive rules on data privacy and prohibits automated decisions without human involvement unless exceptions apply, such as explicit patient consent or justified public interest [92], [93]. By preserving patient rights and promoting confidence in AI-assisted healthcare, these legal protections aim to ensure AI-powered medical imaging methods function within an ethical and legal framework.

Furthermore, for AI to be legal, accountability and transparency must be precisely maintained for the decisions these systems make. In medical imaging, for example, they must produce outputs that are understandable to medical practitioners, allowing for significant human oversight of the AI-enabled decision-making process. When medical professionals can verify AI-driven conclusions, the system's outputs become more credible, allowing for REF-AI integration into diagnostic, prognostic, and therapeutic processes [93], [94]. Moreover, lawful AI systems must account for justice and nondiscrimination to prevent disparities, particularly in diverse healthcare settings and patient populations. Such a commitment to fairness and nondiscrimination is central to lawful AI, as a biased AI algorithm in medical imaging may lead to inconsistent image segmentation, diagnoses, and treatments [29]. Together, these characteristics of lawfulness in AI uphold both the ethical and legal standards essential to patient care, public trust, and positive societal impact [93]–[96].

D. TRACEABILITY

Building REF-AI systems, especially in the high-stakes healthcare industry, requires traceability. Traceability in AI involves maintaining a comprehensive record of each phase in the system's life-cycle, starting from functional and non-functional requirement analysis, system specification, and data collection to AI model development, validation, and deployment [97]. Traceability is an essential safety measure in AI-powered medical imaging, particularly where AI aims at sensitive diagnostic and treatment strategies. It guarantees that any mistakes or discrepancies can be traced to a particular phase of the system's life cycle or an incorrect decision within the AI pipeline, thus promoting adherence to patient safety, clinical standards, and healthcare regulations [98].

To achieve traceability in REF-AI in healthcare, AI systems should be equipped with comprehensive documentation and version control at each stage of development. This includes maintaining detailed records of data sources, core functionalities of the system, training procedures, AI model parameters, and validation strategies [5], [98], [99]. In medical imaging analysis, traceability is particularly critical due to the complex nature of interpreting medical images and the potentially severe consequences of misdiagnosis or misclassification. By tracking each step, from data selection to

automatic image analysis and diagnostic outputs, traceability allows clinicians and regulators to identify the source of errors and any biases in AI recommendations, facilitating accountability and ethical practice in patient care [5], [98]–[101].

E. RELIABILITY

Reliability in REF-AI systems in healthcare means consistently performing as expected across diverse healthcare settings and scenarios, managing uncertainties and errors, and preventing breakdowns or performance issues to maintain core functionalities [102], [103]. In the context of AI, it demonstrates that a system can repeatedly provide accurate and stable outputs, even when faced with unexpected conditions. This is particularly important in medical image analysis, where AI tool sets assist with diagnostics, treatment planning, and patient monitoring and where unreliable outcomes could pose significant clinical risks [97], [104]–[106].

F. EQUITY

Equity in REF-AI, particularly in healthcare, addresses fair access, representation, and outcomes across diverse populations. Equity in AI involves actively mitigating biases that arise from unequal data representation or algorithmic bias. Implementing AI with a foundation of equity is a key component in REF-AI, where studies emphasize that AI systems developed without sufficient regulatory oversight can inadvertently propagate health inequities, negatively affecting marginalized and historically underrepresented groups [107]. In medical imaging, equity is significant as AI-powered models have been used successfully for diagnostics and predictive analysis. Equity-focused AI policies and strategies advocate for regulatory mechanisms requiring diverse data sources and imaging machinery in AI model training. These standards mitigate biases, demonstrating that AI technologies can actively address healthcare disparities [2], [8], [29], [80], [104], [104], [108], [109].

REF-AI offers the potential to advance equity in AI for medical imaging by proposing methods for bias detection and mitigation at every development phase starting from data collection, to AI model development, evaluation, deployment, and interpretation. This includes utilizing federated learning or cyclic weight transfer approaches; allowing institutions to share model knowledge without transferring personal data, thus supporting fair representation while maintaining privacy [108], [110], [111]. By building a regulatory environment prioritizing fairness, REF-AI can act as a revolutionary force in healthcare. This is particularly true in applications such as medical imaging where accurate, reliable, and de-biased decision-making is important for patient outcomes across all populations within different healthcare systems.

G. GOVERNABILITY

Governability in AI is critical to maintaining accountability, upholding ethical standards, and building trust within healthcare systems. The rapid integration of AI in healthcare and

medical imaging analysis begs a robust governance model that addresses ethical and practical challenges [112], [113]. Without clear regulations, AI systems may result in biases that compromise patient privacy and/or diminish clinical confidence. A well-structured and well-organized governance framework is thus needed to build AI applications operating under strict healthcare standards and regulations while being adaptable to emerging challenges. In medical imaging, where accuracy is paramount, governability directly impacts AI-powered diagnostics' reliability, availability, and fairness (e.g., cancerous lung nodules, and bone lucency). Effective governance frameworks reduce risks associated with biased or non-representative images, supporting diagnostic accuracy across diverse patient groups. By embedding these practices, REF-AI in medical imaging can consistently deliver impartial and interpretable results, enhancing diagnostic accuracy and reinforcing trust between patients, clinicians, and healthcare providers [114]. Moreover, a comprehensive governance model that includes inputs from all parties, such as clinicians, AI developers, and patient representatives helps align AI applications with ethical healthcare principles. Mechanisms such as regular audits, performance evaluations across different demographic groups (e.g., sex, race, age, ethnicity), and adherence to privacy make a platform to integrate AI into clinical workflows smoothly. This requires careful attention, as an overly rigid framework here can slow down advancements and innovations in AI integration. At the same time, insufficient oversight may risk patient safety and perhaps ethical breaches [115]–[118].

H. SCALABILITY

Scalability in AI refers to the capacity of AI systems to efficiently handle growing workloads and user numbers and maintain consistent performance as they expand. For medical imaging analysis, scalability is essential due to the growing volume of imaging data and the computational demands of advanced AI methods, such as deep convolutional neural networks (CNNs). The scalability of AI methods now mainly relies heavily on cloud infrastructure, parallel processing, and optimized data management. This collectively enables the high throughput necessary for fast and real-time medical analysis and diagnostics [119]–[122]. Achieving scalable AI in medical imaging involves the adoption of advanced deep learning practices, including transfer learning and federated learning, which allow models to train on varied datasets without centralized data pooling [123]–[126]. In healthcare, AI scalability presents challenges as well. For example, the heterogeneity of medical imaging data requires systems to accommodate different image resolutions, diverse image qualities, and labeling conventions. Cloud-based solutions offer flexible resources and facilitate collaboration across institutions, addressing the high storage and processing requirements while also supporting interoperability across healthcare systems [122].

I. AVAILABILITY

REF-AI should come with AI availability, where it mainly aims to build AI tool sets that are easy to use and accessible for all types of healthcare providers, regardless of technology or data system's level, making them available for all parties [127]–[130]. Different healthcare providers, ranging from big hospitals to smaller clinics, may have vastly different computer systems, equipment, and data types. For AI to be helpful across the board, it must be designed to work well in all these diverse settings. That means building AI tools that are flexible and can also fit smoothly into any healthcare setting: thus, even less-resourced or resource-limited clinics can still benefit from AI-powered technologies.

AI availability also refers to the fact that AI systems can operate across different platforms, such as various operating systems (e.g., Linux, Windows), imaging machines (e.g., MRI, X-ray), and smart devices. This flexibility is needed to meet the demands of diverse clinical environments where technology and equipment can differ significantly. By making AI accessible across these different devices and systems, healthcare organizations can provide clinicians with consistent and reliable AI-powered support in diagnostics, decision-making, and patient monitoring, regardless of their specific tools or machines. Furthermore, the 24/7 availability of AI solutions is crucial in healthcare, as prompt access to AI-driven insights can play a significant impact on patient outcomes during emergencies. Finally, the availability of AI in healthcare is incomplete without addressing AI literacy and education among healthcare professionals. For AI to be successfully adopted, healthcare providers and staff need to understand how to use and benefit from these advanced technologies effectively. This requires investing in AI literacy programs to equip clinicians, technicians, and administrative staff with the knowledge to utilize AI tools confidently. When healthcare providers across the spectrum are trained in AI applications and understand the underlying principles, the sector can move toward a more informed and proactive approach to adopting and implementing AI innovations.

J. EXPLAINABILITY

Explainability in REF-AI clarifies how AI systems make decisions, process data and images, and function internally. By providing transparent explanations of each decision-making step, REF-AI allows users to see exactly how data is processed, what factors contribute to AI-powered decisions, and how results are generated, making AI components more understandable for the end-users (e.g., clinicians, patients, physicians) [104], [131]–[136]. This focus on transparency enables healthcare providers and patients to gain insights into the rationale behind AI-powered decisions and the meaning of these decisions. In this way, REF-AI explainability bridges the gap between complex AI systems and end-users and makes more ethical and informed use of AI, supporting decision-making in high-stakes areas like medical imaging analysis [136]–[141].

In medical imaging, REF-AI's explainability feature is

especially valuable for surgeons, radiologists, and patients, as it helps them understand AI-generated diagnoses or recommendations. This understanding contributes more to shared decision-making, allowing medical professionals to confidently incorporate AI insights while patients feel more informed about their care. By explaining how the AI arrived at each conclusion, REF-AI also reduces the risk of errors or misdiagnosis, while it can also makes these AI-enabled models transparent and free from biases [19], [21], [29], [142]–[147].

K. TRUTHFULNESS

Truthfulness in REF-AI means the ability of AI systems to provide accurate, consistent, and fact-based outputs aligned with medical knowledge and clinical evidence. Medical imaging helps to build AI-powered medical imaging algorithms that are correct and free from misleading or even exaggerated results. Without truthfulness, AI systems risk disseminating inaccuracies that could lead to diagnostic errors (e.g., misclassification), ineffective treatments, prolonged hospitalization, and reduced patient confidence in AI-powered healthcare [67], [142], [148], [149]. AI truthfulness in medical imaging can be illustrated through practical examples. For instance, AI systems that detect cancerous lung nodules from CT images must generate predictions based on verified diagnostic markers, such as size, morphology, texture, or masses, while avoiding false positives caused by image artifacts [150], [151]. Another example is the use of AI for brain tumor segmentation and classification, where truthfulness ensures that highlighted tumor regions correspond accurately to the pathology confirmed through biopsy or clinical evaluation [152]–[154].

Implementing truthfulness in AI systems for medical imaging informatics should include several key elements, including but not limited to (1) rigorous training and validation of AI models using diverse and high-quality datasets to confirm that predictions are based on reliable data representative of real-world scenarios, (2) adherence to medical guidelines and standards, such as clinical protocols, thus it can align AI outputs with currently accepted truths in healthcare, (3) robust error analysis, validation mechanisms, and continuous monitoring for AI-enabled methods, (4) integrating strategies for peer review and clinical oversight to keep AI systems accountable to healthcare providers [149], [155]–[157], and (5) ensure consensus processes that allow incorporation of new knowledge (some generated by widespread use of AI) with consistency across platforms.

L. SAFETY

Safety in REF-AI involves building and implementing AI systems that are robust, dependable, and free from behaviors or predictions that could lead to harm. In medical imaging, where AI supports critical diagnostic and treatment decisions, safety is of paramount importance [112], [158]–[160]. Unsafe AI outputs, such as inaccurate prediction, incorrect tumor segmentation, or misclassification of disease, could

result in delayed treatments, misdiagnoses, or costly and unnecessary procedures. We can protect patients and uphold clinical standards by embedding safety principles into AI systems, enhancing trust among healthcare providers and stakeholders. In medical imaging informatics, for example, AI models for lung nodule detection on CT scans may employ confidence thresholds to identify and triage cases requiring additional human review. Additionally, it can prompt clinicians to cross-reference with other imaging modalities or clinical data, thus reducing the risk of false negative classification.

Regarding safety implementation, robust model validation techniques and testing against diverse and extensive datasets will help consistent performance across various patient demographics and imaging scenarios. Moreover, employing adversarial robustness techniques, such as adversarial training or quantifying errors in predictions, can protect AI systems from malicious inputs that could compromise their accuracy. Continuous monitoring, regular updates, and retraining of AI models on new data can also help maintain safety as clinical practices and/or patient data evolve.

III. REF-AI METHODOLOGIES

This section dissects the methodologies associated with REF-AI frameworks and approaches. Through our analysis of these various methodologies, we aim to emphasize best practices and emerging trends that support effective AI technologies. This review offers a foundational understanding of how these principles can be incorporated into AI systems, enhancing trust and accountability in their applications. Figure 2 illustrates the taxonomy of REF-AI methodologies for medical imaging informatics. Table 1 provides an overview of the key methodologies, strategies, and techniques categorized under the REF-AI framework. The table serves as a concise reference for understanding how REF-AI methodologies contribute to developing equitable, interpretable, and trustworthy AI systems in medical imaging informatics.

A. RESPONSIBLE AI

Responsible AI methods encompass various strategies, including bias detection and mitigation, transparency tools, data privacy, and ethical frameworks, designed to ensure that AI systems are developed and deployed, emphasizing fairness, transparency, and ethical standards, especially in medical imaging. This section discusses methodologies designed to ensure AI responsibility.

1) Bias Detection and Mitigation

Bias mitigation methods are strategies employed first to identify and reduce AI system biases. These biases can arise from data-driven algorithmic and human biases [96], [161]. Ensuring the diversity among patients, stakeholders, healthcare workers, and the datasets used to train and validate machine learning models can help reduce biases and increase equity in outcomes [162], [163]. A list of mechanisms exists to cope with bias detection and mitigation. For instance, causal

models and graphs can detect and address direct discrimination in data, revealing hidden biases and enabling corrective actions [164]. Generating synthetic data helps supplement real-world data, particularly when underrepresented groups lack representation. Resampling techniques, including down-sampling and oversampling, can address imbalances in the training data [165]. Furthermore, careful handling of missing data, particularly for marginalized populations, is essential to avoid introducing further bias, acknowledging that missing data is often non-random and requires thoughtful treatment [80], [164]. From an algorithm perspective, algorithmic adjustments, such as IBM's "adversarial debiasing" technique can act as an adversary by predicting sensitive attributes like race or sex/gender from the data and mitigating their influence on any predictions [163].

Developing solid guidelines emphasizing data origin and quality can help organizations select high-quality datasets. When domain experts and annotators create gold-standard or ground-truth datasets, understanding their experiences and measuring inter-rater agreement becomes important for maintaining data quality and minimizing biases [29], [80], [164].

2) Transparency Mechanisms

Transparency mechanisms are essential in responsible AI to enhance the interpretability and accountability of AI models. Transparency mechanisms will open AI models' black-box nature, offering model-agnostic and model-specific techniques [43]. Such methods could be divided into ex-ante and post-hoc techniques. While the first one incorporates interpretability directly into the AI model during its design and development stage, the latter provides insights into the AI model's process after it has been trained [166], [167]. Ex-ante techniques are proactive and are mainly built into the AI models from the start. Examples include AI models that are intrinsically interpretable, for instance, decision tree models or linear regression models. On the other side, post-hoc methods are applied after the AI models have been trained, so they could make AI predictions more interpretable. These might include visualization methods and feature importance scores that help end-users understand the AI models' decisions. One option would be to provide the positive or negative predictive value of a finding.

3) Ethical Frameworks

Ethical frameworks offer comprehensive guidelines that build fairness and transparency throughout the design, development, and implementation of AI systems. These frameworks facilitate identifying ethical issues, including bias, discrimination, safety, and privacy concerns, enabling healthcare providers to proactively address such issues [168]. The ethical frameworks for responsible AI in healthcare take a multidisciplinary approach, integrating fields such as medical ethics, bioengineering, human well-being, regulatory compliance, and psychology into the AI development process. These frameworks highlight the importance of diverse ex-

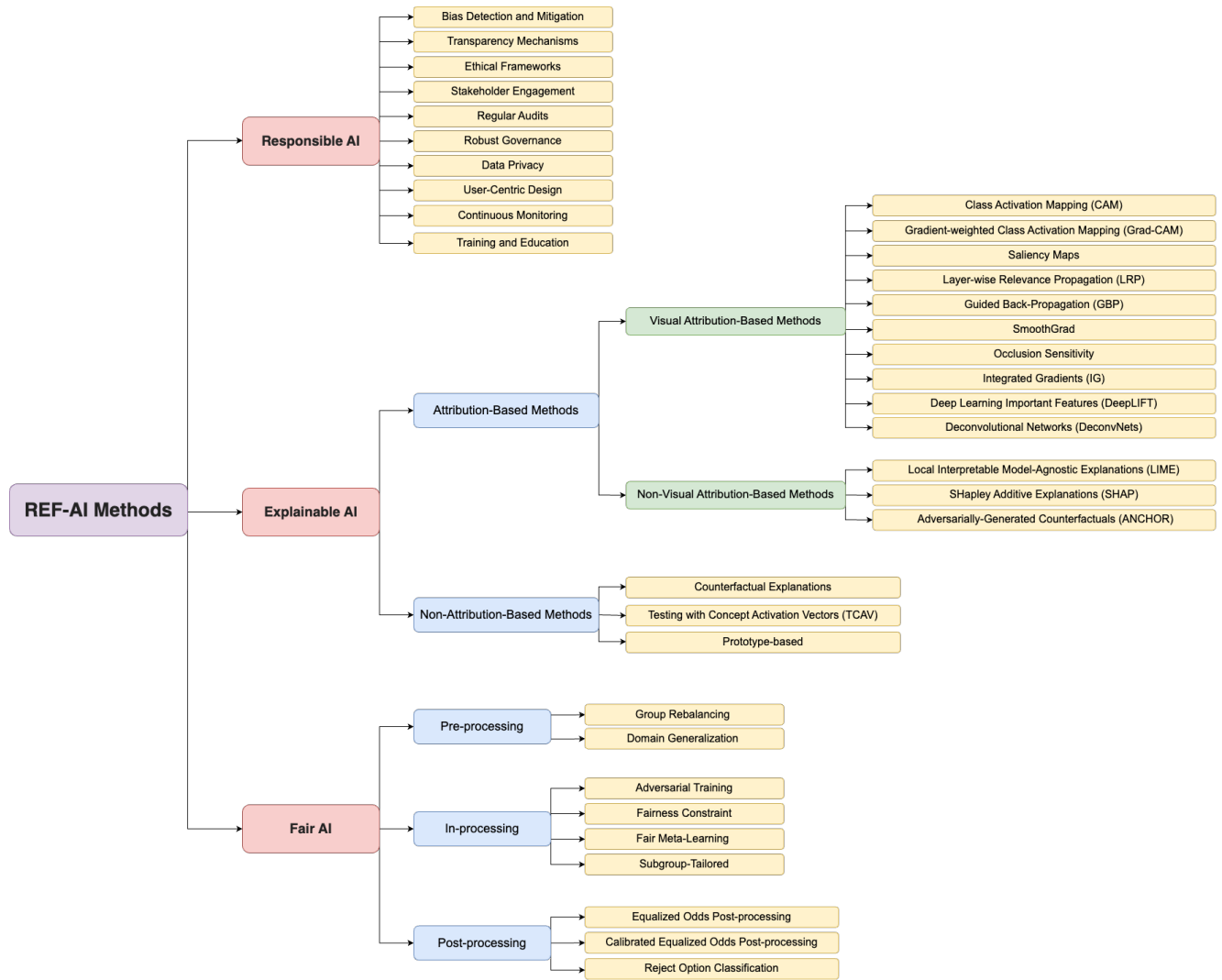


FIGURE 2: The taxonomy of the responsible, explainable, and fair artificial intelligence (REF-AI) methods for medical image analysis.

pertise to not only address but also cope with complex ethical challenges. The responsible design process incorporates ethical impact analysis at each phase, including research, ideation, prototyping, and post-launch evaluation, focusing on psychological well-being and broader ethical issues such as social justice. For example, the “Spheres of Technology Experience” framework organizes the ethical impact of technology across six levels of adoption, interface, task, behavior, life, and society, allowing developers to evaluate and address potential ethical issues at every stage of user interaction and societal influence, making ethical analysis systematic, completely thorough, and actionable [169].

Ethical frameworks in healthcare emphasize the principles of measured action and caring in the in-between to support decision-making that recognizes the interconnectedness of stakeholders and the complexities of healthcare settings. “Measured action” involves taking small, adaptable steps in uncertain situations, while “caring in the in-between”

focuses on making relationships among stakeholders and incorporating their concerns. Together, these principles help ensure the responsible and ethical integration of AI into healthcare, maximizing healthcare benefits while minimizing harm and preventing unequal treatment [96], [166], [168], [170], [171]. Moreover, ethical frameworks promote the protection of privacy by prioritizing patient consent for the use of their protected health information (PHI) and data minimization, as required by several regulations such as GDPR and HIPAA [96].

4) Stakeholder Engagement

Engaging stakeholders is fundamental to the successful establishment of REF-AI. Effective stakeholder engagement brings together AI scientists, healthcare providers, policy-makers, patients, clinicians, health informatics professionals, IT specialists, and experts such as biomedical ethicists. This collaboration makes AI systems align well with ethical and

societal standards, building a safe and sustainable AI ecosystem that benefits all stakeholders, end-users, and society as a whole [2], [100], [163], [165]. Ongoing collaboration among diverse stakeholders, including members of underrepresented and marginalized groups, can help reduce biases and associated risks. This approach leads to more safe and trustworthy AI solutions that address the healthcare community's and society's needs [100], [165].

5) Regular Audits

Regular audits are key components of responsible AI in medical imaging, playing an important role in maintaining ethical standards, detecting and mitigating biases, and evaluating AI system performance against established benchmarks. These audits promote accountability and drive continuous improvement by systematically verifying and validating AI systems for adherence to ethical guidelines, legal frameworks, regulatory standards (e.g., the EU AI Act), and the specific requirements of medical imaging applications [166], [172].

Audits must be integrated throughout the entire AI life cycle, ranging from analysis of requirements and design through development, deployment, and routine utilization. A comprehensive auditing strategy should make the most use of internal and external evaluations conducted at all pre-deployment, post-deployment, and post-incident phases while maintaining audit independence to build an unbiased assessment [100], [172]–[175]. Targeted data audits are essential for systematically evaluating the quantity, diversity, quality, availability, and integrity of data within AI systems. These audits help to first identify and then mitigate risks, such as data silos, biases, and integration challenges. By addressing these issues, data audits enhance the fairness, safety, transparency, and accountability of AI systems in medical imaging, building trust and equity in healthcare outcomes [2], [80], [175], [176].

6) Robust Governance

Robust governance in responsible AI provides a well-organized, structured, and multi-tiered oversight framework that integrates risk management, regulatory compliance, ethical principles, and data governance. By operating across industry, organizational, and team levels, this governance structure helps in implementing AI systems that align well with societal values while advancing accountability, fairness, and transparency throughout their lifecycle [80], [160], [177]–[179]. This includes formal regulations, such as legislative acts and binding guidelines, with voluntary ethical principles to guide the responsible development, deployment, and maintenance of AI systems in healthcare. This dual approach not only enables organizations to achieve their long-term objectives in AI utilization but also safeguards stakeholder interests in AI-powered systems [112], [160], [177], [178], [180].

Effective AI governance, however, integrates several factors, such as risk management, regulatory compliance, and ethical considerations into AI-driven decision-making pro-

cesses to confirm the alignment of the AI methods with societal values. Furthermore, governance structures also emphasize proactive oversight to keep AI-enabled systems accountable and adaptable to contemporary standards and societal expectations [179], [181], [182].

7) Data Privacy

Data privacy is of paramount importance in the development of responsible AI systems. In responsible AI, data privacy includes establishing data governance structures that enforce data lineage, accountability, and adherence to privacy regulations, including HIPAA [183] and GDPR [184]. These frameworks incorporate metadata management through data catalogs and curation, enhancing transparency, integrity, and traceability. Such measures provide compliance with laws and help identify biases, validate data accuracy, and minimize risks associated with data integration, ultimately promoting fairness, accountability, and transparency across AI systems within the healthcare community [80], [185].

Advanced privacy-preserving techniques strengthen AI-powered systems by balancing security, privacy, availability, and efficiency. Federated learning, for instance, enables encrypted model parameters to be shared instead of raw data, maintaining user privacy by keeping data localized [110], [186]. Complementary approaches, such as secure multi-party computation, differential privacy, and trusted execution environments, could provide additional layers of protection [187]–[189]. These techniques facilitate computation on encrypted data instead of real raw and original data, ensuring compliance with privacy regulations (e.g., GDPR) without compromising the accuracy or utility of AI models [186], [190]. Moreover, robust data privacy practices must include informed consent protocols customized to diverse populations. Developing transparent and comprehensive consent is required to have individuals clearly understand how their data will be used, thus empowering them to make informed decisions for their data and how the data will be shared [191], [192].

8) User-Centric Design

User-centric design in responsible AI emphasizes creating AI-powered technologies that prioritize user needs and enhance their overall experience with AI-enabled systems. This approach covers AI fairness, transparency, and ethical use while promoting accessibility and inclusivity for all individuals. By incorporating diverse societal values and adhering to universal design principles, user-centric design facilitates the development of AI systems that align with ethical standards and societal expectations [193]–[195]. In healthcare, a key aspect of user-centric design involves understanding and addressing clinicians' and patients' needs, preferences, and experiences through a focus on the interaction between all end-users and AI-enabled systems. This requires gathering user input, usability analysis, and evaluating systems from subjective perspectives. Different factors, such as AI transparency, fairness, privacy, and explainability are central

to this process. This helps to build and implement AI systems that not only meet technical requirements but also resonate with user expectations and values [193], [195]–[197]. Beyond software and system engineering disciplines, user-centric design incorporates insights from cognitive sciences, psychology, and the humanities to design and develop AI-powered systems that enhance human well-being and align with societal values. This multidisciplinary approach helps to incorporate fairness, transparency, and ethical use while addressing the broader impacts of AI technologies [194].

Moreover, accessibility is another critical pillar of user-centric design. It mainly aims to build AI-enabled systems that accommodate all users, regardless of their age, race, gender, abilities, or characteristics. Adhering to universal design principles may guarantee that AI systems are equitable, inclusive, and accessible to a broad range of end-users. With actively involving diverse cohorts, such as the aging population and individuals with disabilities in the development process, we will be able to implement AI strategies that uphold societal values, promote fairness, and support human dignity [195], [198]–[201].

9) Continuous Monitoring

Continuous monitoring of responsible AI entails the ongoing validation and real-time observation of the functionalities and utilization of AI-enabled systems after deployment. This process maintains adherence to regulations, laws, and ethical standards while providing channels for a diverse range of stakeholders to feedback and report issues and comments [202]–[205]. An important factor for the effectiveness of continuous monitoring is the engagement of a diverse user population, inclusive of different ages, races, genders, abilities, and ethnicities [164].

10) Training and Education

Training and education in responsible AI, delivered through courses, conferences, and multidisciplinary programs provide continuous learning opportunities tailored to diverse audiences. These initiatives equip healthcare professionals, managers, and developers with essential knowledge of AI itself, AI ethics, regulation, clinical applications, and practical tools, thereby advancing trust, safety, and effective implementation of AI in healthcare settings [2], [8], [162], [206], [207]. In addition, training and education methods can improve responsible AI by equipping end-users, such as managers, developers, clinicians, and employees with a strong understanding of AI ethics through structured programs, including but not limited to IEEE's initiative for ethical AI [206], [208], [209].

B. EXPLAINABLE AI

As discussed earlier, explainable AI in healthcare helps to open the black-box nature of AI algorithms and makes it understandable for end-users, including clinicians, physicians, nurses, and patients. Generally speaking, explainable AI methods could be categorized into (1) Attribution-based

and (2) Non-attribution-based methods due to their distinct strategy in interpreting the decision-making processes of AI models (Figure 2). Attribution-based methods focus on quantifying the importance of individual input features, such as specific regions of interest, edges, and/or blobs in a medical image, and provide visual or non-visual representations to highlight these contributions. In contrast, non-attribution-based methods seek to uncover the broader mechanisms and reasoning behind an AI model's predictions or classification, focusing on the overall AI model behavior or high-level summaries that extend beyond individual features [41], [143], [210]. The current section introduces a taxonomy of explainable AI in medical imaging informatics.

1) Attribution-Based Methods

The attribution-based methods in explainable AI are techniques used to enhance the interpretability of AI. The goal of these methods is to determine each feature's contribution to the target output/outcome. Attribution-based methods identify and highlight key features or regions within the input data, providing clearer insights into how the AI model arrives at their decisions [21], [143], [210]–[212]. These methods could then be classified into (1) visual attribution-based methods and (2) non-visual attribution-based methods. While the visual attribution-based methods aim to highlight the regions of the medical image that are contributing significantly to the AI model's prediction, the non-visual attribution-based methods mainly focus on identifying which features or internal model components (e.g., artificial neural network activations) are most responsible for an AI model's decision, without directly highlighting specific regions in the image.

a: Visual Attribution-Based Methods

Class Activation Mapping: Class activation mapping (CAM) [213] is a widely used visualization technique that interprets deep CNNs by highlighting the image regions that are most important for making a specific class prediction. This makes it particularly effective for studies involving medical images [43], [214]–[216]. CAM employs global average pooling (GAP) after the final convolutional layers and before the fully connected layer, computing the class activation map by combining the weights of each convolutional filter with the activations at each spatial location. It highlights the regions most influential in the AI model's decision-making process, thereby visualizing which region of interest in the image is relevant for the AI model's prediction [43], [61]. CAM has been applied in various fields of medical imaging, including cancer identification and tumor classification, by visualizing salient regions in images of the bladder, brain, breast, skin, cardiovascular, chest, gastrointestinal, and thyroid scans [43], [61], [214], [217]–[227].

Gradient-weighted Class Activation Mapping: Gradient-weighted class activation mapping (Grad-CAM) [35], [36] is a local explanation technique that addresses the limitations of CAM. It assigns importance scores to each artificial neuron

by computing gradients flowing into the last convolutional layer, generating a coarse localization map, and highlighting key pixels for class prediction. Grad-CAM calculates artificial neuron importance scores by averaging gradients over the size of the activation map, then combines these weights with forward activation maps to produce a heatmap that highlights areas of the image most relevant for downstream tasks, such as segmentation or classification [42], [43], [53], [59], [62], [64], [228]–[231]. Grad-CAM has diverse applications in medical imaging, such as visualizing regions significant for brain tumor detection, classifying polyps in whole slide images (WSI), identifying glaucoma in OCT scans, or pinpointing decision-critical areas in COVID-19 detection from chest radiographs [43], [61], [232]–[235]. It also enhances visualization for polyp and tumor detection in endoscopic images and aids in gastric cancer classification by highlighting regions essential for diagnostic accuracy [59], [62], [63]. Grad-CAM's utilization extends to breast cancer imaging across different image modalities, such as ultrasound and mammography, where it effectively highlights lesion regions and assists clinicians in understanding any focusing areas [61], [64], [236], [237].

Moreover, Grad-CAM has proven highly effective in various medical imaging applications, enhancing AI model interpretability and diagnosis. It provides accurate heat maps for brain hemorrhage detection, aiding rapid diagnosis of hemorrhage locations [42]. In pneumonia detection, Grad-CAM improves the interpretability of X-ray images by focusing on relevant lung regions, even with background removal [229]. It is also valuable in lung CT imaging for localizing cancerous areas [44] and in brain tumor grading, where it highlights critical features like necrosis [54], [55]. Grad-CAM has been successfully adapted for tumor segmentation in MRI scans, including prostate cancer, offering performance like manual segmentation [230]. Additionally, it enhances radiograph and MRI interpretation in tasks like osteoarthritis diagnosis and tumor segmentation by emphasizing diagnostically relevant features [53].

Saliency Maps: Another mechanism under the category of visual attribution-based methods called saliency maps [238], [239]. Saliency maps are a gradient-based visualization technique that computes the impact of individual pixels on a neural network's classification. They do this by evaluating the gradients of the loss function concerning the input image, revealing which pixels most influence the final decision, and highlighting the relevance of different image areas for a given class [228]. This method highlights critical areas in medical images, helping researchers and clinicians understand where deep learning models focus, and aiding in identifying and diagnosing potential issues or biases. Furthermore, saliency maps are used to explore how demographic factors, such as race and sex affect a deep learning model's predictions of brain regions associated with sex-linked neuropsychiatric conditions. By adding Gaussian noise and averaging the maps over multiple iterations, researchers improved clarity and identified key brain regions associated with different

demographic subgroups [240].

Layer-wise Relevance Propagation: Layer-wise relevance propagation (LRP) [241] is an interpretability method that explains the predictions of neural networks by redistributing the network's output back through the layers to assign relevance scores to individual input features, such as pixels in an image [43], [53], [228], [241]–[243]. Using local redistribution rules, LRP traces how much each artificial neuron contributes to the final output, producing a relevance map that highlights important regions of the input [43], [228], [244]. LRP has been widely used in medical imaging to identify key features for diagnosis. Examples include distinguishing schizophrenia patients using fMRI, creating heatmaps for tumor detection, diagnosing multiple sclerosis, and generating relevance maps for neonatal MRI and Alzheimer's disease detection [43], [243], [244]. Furthermore, LRP is also used to diagnose anterior disc displacement, osteoarthritis, and temporomandibular joint disorder from MRI images, generating heatmaps that serve as visualized rationales for diagnostic predictions [53], [242]. Those studies provide clinicians with interpretable insights and make artificial neural network models more transparent and interpretable for medical diagnosis.

Guided Back-Propagation: Guided back-propagation (GBP) [245] is a visualization and explainability technique that analyzes gradient concerning the input image to highlight those features that are most influential to artificial neuron activation [54], [244]. It changes the backpropagation process by setting gradients to zero for units with zero or negative values after ReLU activation. It highlights features that increase activation and gives a clearer visualization than standard back-propagation [54], [228], [244]. This approach combines ReLU and deconvolution, introducing a guidance signal to prevent the backward flow of negative gradients. It makes it effective for visualizing artificial neural network activations in both the intermediate and final layers [228]. In medical imaging, GBP enables improved visualization and interpretation of diseases using neural networks. It aids in the automated quantification of enlarged perivascular spaces as markers of cerebral small vessel disease in brain MRI. For fMRI, it decodes task states of the human brain without feature engineering. GBP enhances the detection and visualization of bioresorbable scaffolds for coronary heart disease in intravascular optical coherence tomography (IVOCT). In colorectal imagery, it improves semantic segmentation of polyps for cancer prevention with uncertainty estimation. For spinal MRI, it facilitates the grading and localization of pathologies like disc degeneration and stenosis with radiological evidence visualization [246]–[251].

SmoothGrad: SmoothGrad [252] is an enhancement of gradient-based saliency maps that reduces noise by adding Gaussian noise to the input image and averaging the resulting sensitivity maps [43], [252], [253]. It improves the clarity of the saliency map by smoothing gradients, which reveals the effect of small changes in each pixel on the classification score. By applying a Gaussian kernel to average

multiple perturbed images, SmoothGrad refines the saliency visualization and could be combined with other gradient-based methods for better results [43]. In medical imaging, SmoothGrad was applied to breast MRI data to enhance the clarity of feature visualizations generated by the deep CNNs, helping to distinguish relevant spatial and dynamic features from pre-processing artifacts during estrogen receptor status classification [253].

Occlusion Sensitivity: Occlusion sensitivity [238] is an agnostic method that generates saliency maps by systematically occluding parts of the input image and observing changes in the classification score to identify the importance of image regions in a model's decision [214], [238], [254], [255]. In medical image analysis, this technique generates heatmaps highlighting areas that significantly influence the model's decision, as demonstrated in lesion segmentation tasks. The method typically results in "hotter" explanation maps than GradCAM, suggesting more areas are considered highly relevant for segmenting masses [255], [256].

Integrated Gradients: Integrated gradients (IG) [257] is a gradient-based method used to interpret deep learning models by evaluating the contribution of each input feature to the model's prediction. It requires a baseline input, which could be a black/white or a random image, and calculates how the input image's features differ from the baseline to generate an explanatory heatmap [43], [242], [258]. IG satisfies the axioms of sensitivity and implementation invariance, making it a powerful tool for understanding feature importance and data skew [43], [257]. IG has been used in medical imaging to generate heatmaps for predicting diabetic retinopathy severity, providing pixel-level insights into feature contributions [259]. It has also visualized features in a deep CNN trained to classify estrogen receptor status from breast MRI [253] and helped create explainable heatmaps for diagnosing temporomandibular joint disc displacement using MRI images, supporting clinical decision-making [242].

Deep Learning Important Features: Deep learning important features (DeepLift) [260] is a back-propagation-based method that calculates contribution scores, such as LRP and IG by comparing changes in artificial neuron activations between an input and a reference. By propagating these differences through the network, DeepLift quantifies how every feature contributes to the final prediction [43], [228]. It addresses issues with gradient-based methods, such as gradient zeroing and discontinuities, allowing for a more reliable interpretation of model decisions [43]. AI-powered Medical image analysis uses DeepLift to identify the salient features for Multiple Sclerosis classification, due to its quantitative evaluation performance among other visually explainable AI methods [228], [261], [262].

Deconvolutional Networks: Deconvolutional networks (DeconvNets) [238] is an explainability technique used to visualize activations in CNNs by mapping pixel-level learning back to the input layer. This method constructs a deconvolution network by adding transposed convolution and unpooling layers to reverse the effects of convolution and

max pooling layers in the original model. Deconvnets can effectively visualize activations for convolution and max pooling layers by generating visualizations that map learned features back to the input pixels [44].

b: Non-Visual Attribution-Based Methods

Local Interpretable Model-Agnostic Explanation: Local interpretable model-agnostic explanation (LIME) [263] perturbs the original images, feeds them into the black-box model, and examines the resulting outputs. It then assigns weights to these perturbed images based on their proximity to the original images [43], [56], [214], [231], [263]. It then fits a surrogate computational model, such as linear regression, to the weighted dataset of perturbed points, explaining the original image's prediction. LIME is applied to brain MRI data to reveal visual evidence supporting Alzheimer's disease classification using deep CNNs, and provides interpretable insights for medical professionals by pinpointing key brain regions that influence predictions [65], [264].

LIME generates heat maps and super-pixels to highlight key features influencing classification decisions for COVID-19 and pneumonia in CT and X-ray images, enhancing AI model explainability and aiding clinicians in understanding the decision-making process [265], [266]. It was also used to interpret deep learning predictions for retinoblastoma, identifying important regions in fundus images, improving transparency and detection accuracy [56]. Additionally, LIME provided plausible explanations for an endoscopic dataset including Lymphangiectasia and Pylorus using endoscopic images [231], and interpreted MRI patches for glioblastoma multiforme detection [267]. LIME has also improved the explainability of cancer detection models, such as those for colorectal cancer and osteosarcoma, by balancing accuracy with interpretability using histopathological images [268]. It further offers valuable visual explanations for CNN models in digital tomosynthesis images of breast lesion classification, highlighting influential regions [64].

SHapley Additive Explanations: SHapley Additive Explanations (SHAP) [269] is a game-theory strategy [270], [271] utilized to explain predictions by attributing the contribution of every input feature to the overall prediction of the AI model [56], [231], [272]. SHAP ensures local accuracy by aligning predictions with the expected average for simplified inputs, missingness by excluding features absent from the original input, and consistency by increasing the contribution of simplified inputs and the SHAP value if the model changes. This guarantees reliable feature attribution in AI model explanations [43]. In medical imaging, SHAP has been applied to analyze the output of a 3D regression CNN for estimating volumetric breast density from MRI images, identifying key features and highlighting inaccuracies related to structures like the pectoral muscle and heart, thus confirming the feasibility of estimating breast density without segmentation [272]. It was also used to enhance interpretability for retinoblastoma images by assigning importance scores to pixels, identifying key regions such as yellow-white masses

and calcifications, and revealing the absence of critical features in normal images [56]. Additionally, SHAP was also employed to explain an AI model for early detection of mutations in the Kirsten Rat Sarcoma viral oncogene homolog (KRAS) and epidermal growth factor receptor (EGFR) in lung cancer patients using low-dose computed tomography (LDCT) images. [273].

Adversarially-Generated Counterfactuals: Adversarially-generated counterfactuals (ANCHOR) [274] is a method that provides stable explanations for model predictions by using a set of if-then rules to anchor a prediction to identify crucial features or segments in an image which consistently lead to a particular prediction. This method ensures that the variation among the remaining segments does not influence the prediction. In medical imaging, ANCHOR can generate a fairly intuitive explanation for a chest X-ray classification model for identifying COVID-19 patients by providing most of the left side of the lungs as the explanation for the prediction [43], [275].

2) Non-Attribution-Based Methods

Non-attribution-based methods do not directly attribute importance to specific parts of the image; instead, they aim to reveal the underlying processes and rationale behind an AI model's predictions, offering explanations that go beyond pixel-level analysis [41], [210].

Counterfactual Explanations: Counterfactual explanations [276] are commonly employed in explainable AI to provide "what-if" insights, which explore how changes to the input data would affect the AI model's prediction. These explanations involve making minimal alterations to an image to interpret the AI model's decision for an individual instance [43]. In medical imaging, counterfactual explanations modify or exclude specific regions, such as pathological areas, to observe how predictions change. This approach helps identify key areas that influence the model's decision while preserving the interpretability of the model [277].

Testing with Concept Activation Vectors: Testing with concept activation vectors (TCAV) [278] is a non-visual global method in explainable AI that quantifies how high-level concepts, such as specific features like a tumor in a medical image, influence an AI model's predictions or decisions [43], [61], [279]. Several studies have applied TCAV to medical image analysis, including diabetic retinopathy (DR), cardiac MRI, breast cancer detection, and skin lesion classification [43], [278], [280]–[283]. More specifically, TCAV was used to identify significant diagnostic features like microaneurysms or aneurysms for DR and to assess clinically meaningful biomarkers, such as ventricular ejection in cardiac MRI [278], [280], [281]. Additionally, TCAV has been extended to regression problems and used in skin lesion classification to highlight important dermoscopic features [43], [279]. TCAV has also been used to evaluate how features, such as temperature gradients or vascular structures affect classification decisions in infrared breast images [61]. TCAV with a discovering phase enhances AI interpretability

in cardiac MRI by identifying key features linked to cardiac conditions, providing clinically meaningful explanations and a quantitative measure of feature importance while reducing pre-processing time [280].

Prototype-based: Prototype-based methods [284]–[287] in AI involve representing data points using a set of prototype examples that capture the essence of different classes or decisions. Each prototype serves as a reference point for classifying new samples mainly based on their similarity to these prototypes. Prototype-based methods involve using representative examples or prototypes to explain and interpret model predictions [288]. In medical imaging, studies have used prototype-based methods for various applications including lesion classification, thyroid nodule diagnosis, cancer detection, and COVID-19 detection from chest X-rays [43]. In addition, ProtoTree, a Prototype-based method, is used in infrared breast image classification by employing a decision tree structure where each leaf represents a prototype for a class of images and classifies images based on their proximity to these prototypes [61]. Techniques, such as influence functions and variational autoencoders (VAEs) have been used to identify significant features, cluster image patches, and generate visual interpretations of prototypes, aiding in understanding and explaining model predictions and improving diagnostic accuracy [43].

C. FAIR AI

Fairness in AI could be classified into three different strategies, including pre-processing, in-processing, and post-processing strategies. These techniques are mainly based on when and how bias detection and mitigation techniques intervene in the AI model pipeline (Figure 2). Pre-processing methods involve organizing or adjusting the input data to mitigate bias before training an AI model. Common methods include data reweighting, which assigns different weights to samples to correct imbalances, and data anonymization, which removes sensitive attributes to prevent discrimination based on protected characteristics. Other methods, such as data resampling and adversarial debiasing are also employed to ensure fairness in the model's performance [32], [60], [289]–[294]. In contrast, in-processing strategies, such as adversarial training and fairness constraint methods, intervene during model training to mitigate bias and enforce fairness [32]. Post-processing strategies, on the other hand, adjust model outputs to calibrate predictions and ensure fairness across sub-groups [60].

1) Pre-processing

Group Rebalancing: Group rebalancing is a technique to address class or group bias in datasets to ensure that each group or class is represented more equally, improving model fairness and performance across diverse categories. Group rebalancing typically involves techniques, such as data resampling [290], data reweighting [295], stratified batching [296], and data augmentation [297]. Data resampling involves either oversampling minorities or undersampling majorities in

order to ensure a balanced representation across the classes. In cardiac MR image analysis, the data resampling methods have been used to ensure that each batch of data includes an equal representation of all protected groups, thereby enhancing fairness in the AI model's training process [60], [298]. Data resampling and stratified batching have also been employed to address biases related to gender, race, and age for deep learning-based segmentation of the skeletal anatomy of the knee and hip joints in plain radiographs [29]. Data reweighting could cope with addressing label noise and imbalanced datasets in medical images by giving individual weights to training samples to reduce bias [294]. In medical images, data reweighting can improve model performance on noisy labeled data, such as skin lesion datasets, and can be applied to various medical image classification tasks without requiring pre-estimated noise distributions or clean data [294]. Data augmentation artificially boosts the diversity of training data (e.g., images) by generating variations of existing images, such as through rotations, flips, or noise addition [299]. Data augmentation has effectively reduced diagnostic accuracy disparities, such as those in diabetic retinopathy between different skin tones [32]. Moreover, the data augmentation method alters the retinal appearance and diabetic retinopathy status to address imbalances related to skin color in retinal images and improve fairness in medical image analysis [300]. It can also improve the performance and fairness of classifiers in histopathology, chest X-ray, and dermatology, especially for underrepresented groups and out-of-distribution cases [299].

Domain Generalization: Domain generalization (DG) [301] aims to maintain good performance across diverse and unseen sub-populations by addressing distribution shifts. DG methods include Group Distributionally Robust Optimization (GroupDRO), which reduces worst-case loss through stronger regularization, and Stochastic Weight Averaging Densely (SWAD), which improves model performance by finding robust flat minima through dense weight sampling. Another method, Sharpness-Aware Minimization (SAM), focuses on parameters in regions of consistently low loss to enhance generalization [298]. In medical imaging, DG is crucial because AI models trained on data from specific hospitals or populations might not generalize effectively to data from different sources or patient demographics. DG techniques aim to improve model robustness and fairness by minimizing domain-specific biases and ensuring that the learned features represent the underlying medical conditions rather than the characteristics of a particular dataset.

2) In-processing

Adversarial Training: Adversarial training [302] is a computational strategy to address biases by enhancing the primary model's performance on the target variable while preventing a secondary model from predicting sensitive attributes based on the primary model's features [32], [298]. Adversarial training has been shown to effectively reduce biases in skin lesion classification [32]. Adversarial training

was also employed to predict HIV diagnosis from MRI scans, identify sex differences in adolescents using data from the National Consortium on Alcohol and Neurodevelopment in Adolescence (NCANDA), and identify bone age from plain radiographs [303]. This method can achieve accurate predictions while reducing biases related to confounder [303].

Fairness Constraint: Fairness constraint [304] is a mechanism incorporated into machine learning models to ensure that predictions are not influenced by sensitive attributes in the data, such as gender or race [305]. In medical imaging, fairness constraints can be applied to prevent disparities in diagnostic accuracy based on protected attributes, such as age, gender, race, and ethnicity to ensure equitable treatment across different patient demographic groups. More specifically, the fairness constraint has been utilized in artificial neural network architectures for dermatology medical image analysis, which involves the introduction of an unfairness score, which is the difference in precision between light and dark skin datasets, thus it could help to minimize this score through targeted fairness constraints [306]. The fairness constraint is applied to a multiexit convolutional neural network (ME-CNN) to achieve fairness in the diagnosis of dermatological disease without using sensitive attributes. This addresses concerns about privacy and availability while improving discrimination based on low-level features and optimizing the accuracy and fairness balance for each test instance [307].

Fair Meta-Learning: Fair meta-learning is a method that trains a model to optimize both accuracy and fairness by introducing a meta-fair classifier into an AI-powered model [60]. For example, in tasks like cardiac MR image segmentation, this method has been used to handle image segmentation and classification of protected attributes in a multi-task learning framework. This approach aims to ensure that no single group disproportionately affects the learning of the model, thus promoting fairness between different groups of patients in medical imaging [60].

Subgroup-Tailored: Subgroup-tailored modeling [29], [308], [309] enhances model performance for minority groups by training individual AI models specific to each group, enabling evaluation of group-specific AI models' effectiveness in mitigating bias and improving overall fairness. In medical imaging, the subgroup-tailored modeling method, applied to automated image segmentation of knee and hip anatomy using plain radiographs, improved fairness by reducing racial biases [29]. This tailored strategy ensures that the AI model(s) performs well in diverse populations, addressing potential disparities, and improving the general fairness in medical image analysis.

3) Post-processing

Equalized Odds Post-processing: Equalized odds post-processing (EOP) [310] corrects the output of an existing AI algorithm to satisfy equalized odds, a fairness measure requiring privileged and unprivileged groups to have identical false positive and false negative rates. It happens by solving a

linear program to adjust output labels probabilistically [60], [72], [311]. EOP is applied to achieve group fairness in chest X-ray classifiers [66].

Calibrated Equalized Odds Post-processing: Calibrated equalized odds post-processing [312] builds upon the equalized odds approach. However, it adjusts the output labels by optimizing classifier scores to probabilistically meet the equalized odds requirement. It mainly focuses on ensuring that the adjusted predictions are both well-calibrated and fair [60], [72], [311].

Reject Option Classification: Reject option classification [66] aims to reduce bias by adjusting predictions for cases where the AI model is uncertain. It works by giving preference to unprivileged groups and reducing the advantages for privileged groups within a specific confidence range around the decision boundary. This method uses probabilistic classifiers or ensemble classifier disagreements to make decisions in high-uncertainty situations without changing the original data or model. It also allows for flexible control over fairness, and it can handle multiple sensitive attributes at the same time [60], [72], [311].

IV. REF-AI EVALUATION METRICS

This section provides a brief overview of the evaluation metrics available for REF-AI.

A. RESPONSIBLE AI

Evaluating responsible AI requires qualitative and quantitative approaches, focusing on societal impacts, stakeholder engagement, robust governance, and adherence to ethical principles. This requires conducting iterative assessments within the AI lifecycle, ranging from requirement analysis to development and deployment. These assessments should integrate qualitative feedback from diverse groups, including clinicians, patients, policymakers, and marginalized communities, with quantitative analysis. In addition, methods such as surveys, participatory design evaluations, and ethical impact assessments are employed to ensure comprehensive and inclusive evaluation [2], [100], [313]–[317].

Unlike explainable and fair AI, which benefits from well-established metrics and measurement approaches, responsible AI lacks widely accepted standardized metrics. Instead, it relies on qualitative evaluations and emerging frameworks that address diverse aspects such as accountability, transparency, robust governance, data privacy, and ethical compliance. The challenge lies in developing cohesive standards to measure the subjective dimensions of responsibility in AI [2], [8], [166], [178], [318]–[321].

B. EXPLAINABLE AI

Subjective and objective metrics provide different approaches for evaluating AI explanations, each with its own methods and goals. Together, these metrics ensure human interpretability and AI model transparency (Figure 3). Here, we briefly discuss these metrics.

1) Subjective Metrics

Subjective metrics include human evaluation to assess the quality and accuracy of AI explanations. These metrics rely on human judgment to assess how well an AI explanation is understood and how it influences trust among AI and end-users. This process often requires expert insights to determine if explanations align with human understanding and expectations. These metrics are often gathered through user studies or surveys, with evaluators assessing the relevance and correctness of AI explanations to provide more precise insights into the model's behavior and its decision-making processes [43], [322], [323].

2) Objective Metrics

In contrast, objective metrics focus on measurable criteria (e.g., fidelity and consistency). These objective metrics in AI explainability could be divided into model-based and specific metrics.

Model-based Metrics: These metrics focus on directly interacting with an AI model's predictions or processes, providing clear and measurable evaluations. Attribution-based metrics could check how well an AI explanation matches the AI model's importance assigned to different features. For example, they might measure the accuracy of saliency maps, the relevance of highlighted areas, or correlation and/or IoU among areas automatically detected by AI and those manually provided by the domain experts. Perturbation-based metrics look at how changes to the input or features affect the model's predictions, helping to understand the stability and reliability of the explanations. Additionally, model performance metrics assess how well the explanations reflect the model's behavior, often by tracking changes in accuracy or consistency when the inputs are altered [324]–[329].

- **Attribution-Based Metrics:** To comprehensively evaluate attribution-based metrics, various methods could assess how accurately an AI explanation highlights relevant features so that key regions influencing model predictions are correctly identified and aligned with ground truth areas. These metrics could be categorized as follows.

Pointing Game Metrics: Pointing game metrics evaluate the precision of the saliency maps in identifying the relevant regions [43], [214], [330]. Pointing game metric measures how well the maximum point from the saliency map aligns with the bounding box of a specific class. A hit is counted if the point falls within the bounding box; otherwise, it's a miss, and the accuracy is obtained from the number of misses and hits for each object [214].

Attribution Localisation Metrics: Attribution Localisation metrics [331] are calculated as the ratio of the sum of positive attributions within the bounding box to the sum of overall attributions [332], [333].

Top-k Intersection: Top-k intersection [334] evaluates the consistency of feature importance between different images. The top-k intersection measures the overlap of

TABLE 1: An overview of the key methodologies, strategies, and techniques categorized under the REF-AI framework.

Category	Method(s)	Reference(s)
Responsible AI	Bias Detection and Mitigation	(Norori et al., 2021) [161], (Sargent et al., 2021) [162], (Schwartz et al., 2022) [164], (Werder et al., 2022) [80], (Mensah et al., 2023) [163], (Cary et al., 2024) [165], (Siddiqui et al., 2024) [29], (Nasir et al., 2024) [96]
	Transparency Mechanisms	(Szabo et al., 2022) [167], (Diaz et al., 2023) [166], (Hossain et al., 2023) [43]
	Ethical Frameworks	(Peters et al., 2020) [169], (Li et al., 2022) [168], (Diaz et al., 2023) [166], (Valles et al., 2023) [170], (Bekbolatova et al., 2024) [171], (Nasir et al., 2024) [96]
	Stakeholder Engagement	(Siala et al., 2022) [2], (Gkontra et al., 2023) [100], (Mensah et al., 2023) [163], (Cary et al., 2024) [165]
	Regular Audits	(Werder et al., 2022) [80], (Siala et al., 2022) [2], (Diaz et al., 2023) [166], (Gkontra et al., 2023) [100], (Xia et al., 2024) [172], (Esmailzadeh et al., 2024) [176], (Li et al., 2024) [173], (Murikah et al., 2024) [174], (Riemer et al., 2024) [175]
	Robust Governance	(Macrae et al., 2019) [160], (Ho et al., 2019) [180], (Guan et al., 2019) [182], (Reddy et al., 2020) [112], (Khanna et al., 2021) [181], (Werder et al., 2022) [80], (Camilleri et al., 2024) [177], (Ligot et al., 2024) [179], (Lu et al., 2024) [178]
	Data Privacy	(Yao et al., 1982) [187], (Garfinkel et al., 2003) [189], (Dwork et al., 2008) [188], (Yang et al., 2021) [186], (Ng et al., 2021) [110], (Kotsenas et al., 2021) [192], (Werder et al., 2022) [80], (morley et al., 2022) [185], (Rudd et al., 2023) [190], (Hurley et al., 2024) [191]
	User-Centric Design	(Dignum et al., 2019) [194], (Nevanpera et al., 2021) [195], (Oberste et al., 2022) [196], (H et al., 2023) [197], (Torkamaan et al., 2024) [193]
	Continuous Monitoring	(Nurhaliza et al., 2020) [205], (Sanderson et al., 2022) [203], (Lu et al., 2022) [204], (Schwartz et al., 2022) [164], (Radanliev et al., 2024) [202]
	Training and Education	(Chatila et al., 2017) [209], (Economou et al., 2019) [208], (Sargent et al., 2021) [162], (Walsh et al., 2023) [8], (Wang et al., 2020) [206], (Siala et al., 2022) [2], (Dominguez et al., 2023) [207]
Explainable AI	Class Activation Mapping (CAM)	(Zhou et al., 2016) [213], (Rajpurkar et al., 2018) [227], (Ausawalaithong et al., 2018) [224], (Shinde et al., 2019) [217], (Gao et al., 2019) [218], (Lee et al., 2019) [219], (Lee et al., 2019) [220], (Dunmon et al., 2019) [225], (Luo et al., 2019) [222], (Lei et al., 2020) [221], (Wang et al., 2020) [223], (Li et al., 2020) [226], (Jung et al., 2021) [216], (Miro et al., 2022) [214], (Hossain et al., 2023) [43], (Raghavan et al., 2024) [61], (Wang et al., 2024) [215]
	Gradient-weighted Class Activation Mapping (Grad-CAM)	(Korbar et al., 2017) [232], (Selvaraju et al., 2017) [35], (Pereira et al., 2018) [54], (Thakoor et al., 2019) [234], (Lin et al., 2020) [235], (Vuppala et al., 2020) [44], (Windisch et al., 2020) [233], (Kim et al., 2021) [42], (Saleem et al., 2021) [59], (Karim et al., 2021) [53], (Zhang et al., 2021) [36], (Esmaili et al., 2021) [55], (Javali et al., 2022) [63], (Yang et al., 2022) [229], (Hussain et al., 2022) [64], (Gunasekar et al., 2022) [230], (Mukhtarov et al., 2023) [62], (Varam et al., 2023) [231], (Hossain et al., 2023) [43], (Raghavan et al., 2024) [61], (Lai et al., 2024) [228], (Kajala et al., 2024) [236], (Talaat et al., 2024) [237]
	Saliency Maps	(Simonyan et al., 2013) [239], (Zeiler et al., 2014) [238], (Stanley et al., 2022) [240], (Lai et al., 2024) [228]
	Layer-wise Relevance Propagation (LRP)	(Bach et al., 2015) [241], (Bohle et al., 2019) [244], (Yoon et al., 2023) [242], (Shin et al., 2023) [243], (Karim et al., 2021) [53], (Hossain et al., 2023) [43], (Lai et al., 2024) [228]
	Guided Back-Propagation (GBP)	(Springenberg et al., 2014) [245], (Jamaludin et al., 2017) [251], (Pereira et al., 2018) [54], (Dubost et al., 2019) [247], (Bohle et al., 2019) [244], (Eitel et al., 2019) [246], (Gessert et al., 2019) [249], (Wickstrom et al., 2020) [250], (Wang et al., 2020) [248], (Lai et al., 2024) [228]
	SmoothGrad	(Smilkov et al., 2017) [252], (Papanastopoulos et al., 2020) [253], (Hossain et al., 2023) [43]
	Occlusion Sensitivity	(Zeiler et al., 2014) [238], (Fong et al., 2017) [254] (Miro et al., 2022) [214], (Farrag et al., 2023) [255], (Chen et al., 2024) [256]
	Integrated Gradients (IG)	(Sundararajan et al., 2017) [257], (Ancona et al., 2017) [258], (Sayres et al., 2019) [259], (Papanastopoulos et al., 2020) [253], (Yoon et al., 2023) [242], (Hossain et al., 2023) [43]
	Deep Learning Important Features (DeepLIFT)	(Shrikumar et al., 2017) [260], (Hossain et al., 2023) [43], (Jin et al., 2023) [262], (Patil et al., 2024) [261], (Lai et al., 2024) [228]
	Deconvolutional Networks (DeconvNets)	(Zeiler et al., 2014) [238], (Vuppala et al., 2020) [44]
	Local Interpretable Model-Agnostic Explanation (LIME)	(Ribeiro et al., 2016) [263], (Rucco et al., 2020) [267], (Ahsan et al., 2021) [266], (Kamal et al., 2021) [264], (Hossain et al., 2022) [64], (Miro et al., 2022) [214], (Hossain et al., 2023) [43], (Duamwan et al., 2023) [65], (Ghnemat et al., 2023) [265], (Varam et al., 2023) [231], (Aldughayfiq et al., 2023) [56], (Alkhalaf et al., 2023) [268]
	SHapley Additive Explanations (SHAP)	(Von et al., 1947) [271], (Fudenberg et al., 1991) [270], (Lundberg et al., 2017) [269], (Van et al., 2020) [272], (Le et al., 2021) [273], (Hossain et al., 2023) [43], (Aldughayfiq et al., 2023) [56], (Varam et al., 2023) [231]
	Adversarially-Generated Counterfactuals (ANCHOR)	(Ribeiro et al., 2018) [274] (Abeyagunasekera et al., 2022) [275], (Hossain et al., 2023) [43]
	Counterfactual Explanations	(Wachter et al., 2017) [276], (Lenis et al., 2020) [277], (Hossain et al., 2023) [43]
	Testing with Concept Activation Vectors (TCAV)	(Kim et al., 2018) [278], (Clough et al., 2019) [281], (Lucieri et al., 2020) [283], (Tjoa et al., 2020) [279], (Gamble et al., 2021) [282], (Janik et al., 2021) [280], (Hossain et al., 2023) [43], (Raghavan et al., 2024) [61]
	Prototype-based	(Li et al., 2018) [284], (Chen et al., 2019) [285] (Donnelly et al., 2022) [286], (Hossain et al., 2023) [43], (Bodria et al., 2023) [288] (Rath et al., 2024) [287], (Raghavan et al., 2024) [61]
	Group Rebalancing	(Chawla et al., 2002) [290], (Kamiran et al., 2012) [295], (Xue et al., 2019) [294], (Shorten et al., 2019) [297], (Burlina et al., 2021) [300], (Puyol et al., 2021) [60], (Zong et al., 2022) [298], (Ricci et al., 2022) [32], (Afzal et al., 2023) [296], (Siddiqui et al., 2024) [29], (Ktena et al., 2024) [299]
Fair AI	Domain Generalization	(Blanchard et al., 2011) [301], (Zong et al., 2022) [298]
	Adversarial Training	(Goodfellow et al., 2020) [302], (Zhao et al., 2020) [303], (Ricci et al., 2022) [32], (Zong et al., 2022) [298]
	Fairness Constraint	(Dwork et al., 2012) [304], (Zafar et al., 2019) [305], (Yang et al., 2023) [306], (Chiu et al., 2023) [307]
	Fair Meta-Learning	(Puyol et al., 2021) [60]
	Subgroup-Tailored	(Kearns et al., 2018) [308], (Feuerriegel et al., 2020) [309], (Puyol et al., 2021) [60], (Siddiqui et al., 2024) [29]
	Equalized Odds Post-processing	(Hardt et al., 2016) [310], (Lohia et al., 2019) [311], (Puyol et al., 2021) [60], (Zhang et al., 2022) [66], (Ferrara et al., 2023) [72]
	Calibrated Equalized Odds Post-processing	(Pleiss et al., 2017) [312], (Lohia et al., 2019) [311], (Puyol et al., 2021) [60], (Ferrara et al., 2023) [72]
	Reject Option Classification	(Lohia et al., 2019) [311], (Puyol et al., 2021) [60], (Zhang et al., 2022) [66], (Ferrara et al., 2023) [72]

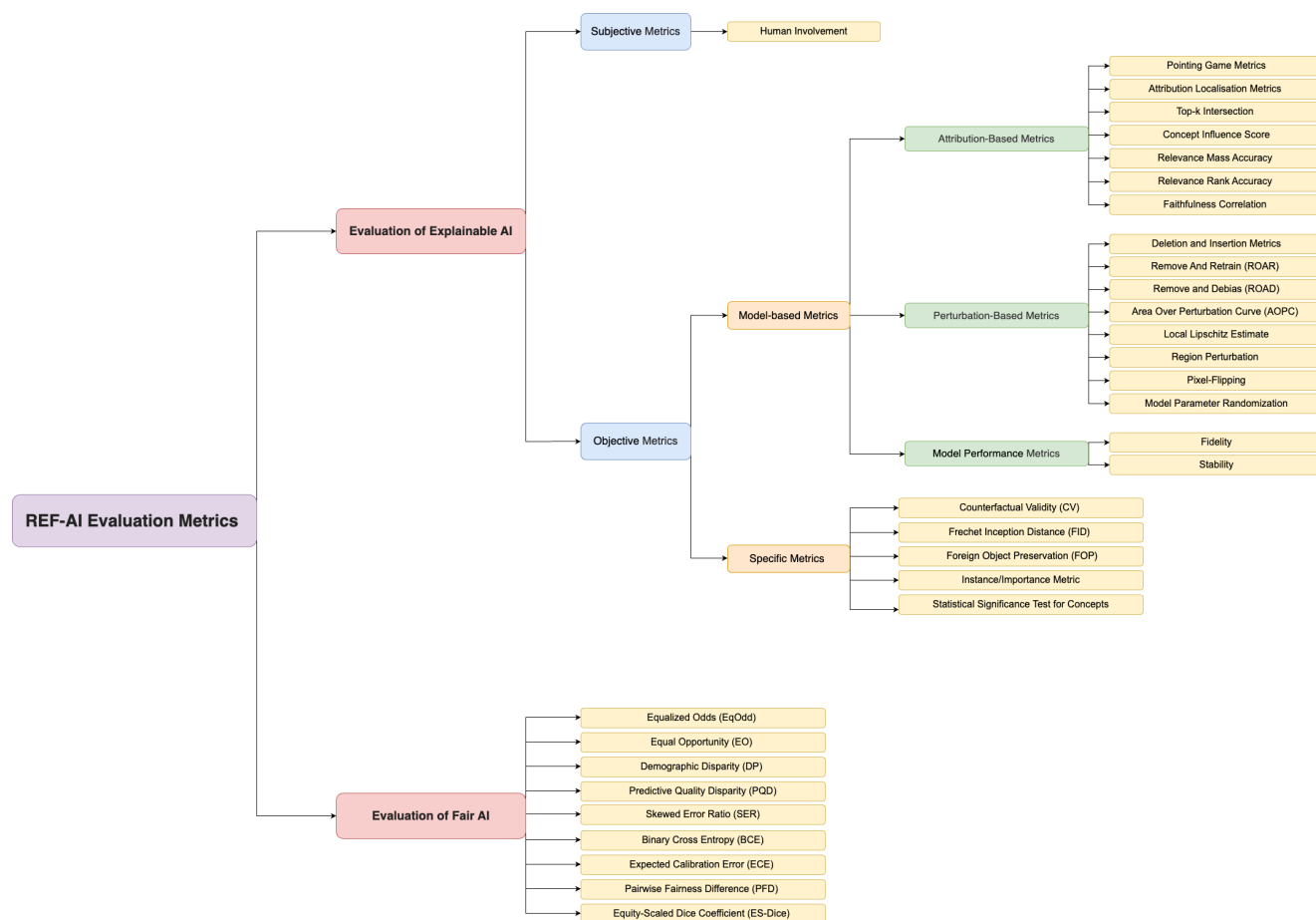


FIGURE 3: The taxonomy of the responsible, explainable, and fair artificial intelligence (REF-AI) evaluation metrics for medical image analysis.

the top-k most relevant features between the original and perturbed image [43], [334].

Concept Influence Score: This score assesses the relevance of high-level visual concepts in AI model predictions. The concept influence score measures the pixel-wise intersection between an explanation and a segmentation map, focusing on semantic concepts influencing predictions [43], [335].

Relevance Mass Accuracy: Relevance mass accuracy [336] evaluates how well the ground truth mask captures the relevant features. It is calculated as the ratio of the sum of positive relevance values within the bounding box to the sum of all positive attributions in the image [43], [336]–[338].

Relevance Rank Accuracy: Relevance rank accuracy [336] assesses the concentration of essential features within the relevant areas. It determines how high-intensity relevance is included within the ground truth mask [43], [336], [338], [339].

Faithfulness Correlation: Faithfulness correlation [340] evaluates the faithfulness of the explanation to the model's predictions and measures the correlation between the explanation and the actual model behavior

for [43], [336]–[339], [341].

- **Perturbation-Based Metrics:** Perturbation-based metrics assess how changes in features affect AI model predictions. By modifying specific input areas, pixels, or model parameters, these metrics test the stability and consistency of AI explanations by observing how the output and explanations change. These metrics could be classified as follows.

Deletion and Insertion Metrics: Deletion and insertion metrics [342] examine the effect of removing or adding important pixels to observe changes in class probability. Deletion measures class probability decrease, when key pixels are removed, while insertion tracks probability increase with added pixels [43], [59], [342]–[344].

Remove And Retrain: Remove and retrain (ROAR) [345] determines the effect of feature removal on model performance. ROAR involves perturbing the highest-scoring regions, retraining the AI model on these perturbed images, and checking for changes in accuracy [343], [344], [346].

Remove and Debias: Remove and debias (ROAD) [347] reduces computational costs while evaluating feature

importance. ROAD measures the impact of removing features without retraining, using mutual information to assess the contribution of low-important pixels [43], [347].

Area Over Perturbation Curve: Area Over Perturbation Curve (AOPC) [342] quantifies the impact of perturbations on the saliency map's relevance. It measures the difference in certainty of object presence with and without perturbations and higher AOPC values indicate greater feature relevance [348]–[350].

Local Lipschitz Estimate: Local Lipschitz [351] Estimate assesses the stability of explanations across similar inputs and evaluates the consistency of explanations for similar instances using Lipschitz continuity [351], [352].

Region Perturbation: Region perturbation [342] measures the contribution of different regions to the model's output, and it perturbs specific regions in the input to evaluate their impact on the model's predictions [342], [353]–[356].

Pixel-Flipping: Pixel-flipping [241] assesses the importance of specific pixels in the explanation and it tests how flipping individual pixels affects the model's predictions [255], [357], [358].

Model Parameter Randomization: Model parameter randomization [359] assesses how explanations hold up under changes and it evaluates the robustness of the explanation by randomizing model parameters and checking for consistency [43], [59], [230].

- **Model Performance Metrics:** Model performance metrics offer essential insights into the reliability and transparency of AI model explanations by evaluating how well these explanations align with the model's decision-making and maintain consistency under similar conditions. This includes two metrics as follows.

Fidelity: Fidelity [360] measures how closely the methods resemble or mimic the decision-making process of the AI model, where high fidelity means that the AI explanation is a true representation of how the model makes its decisions [344], [360]–[363].

Stability: Stability [364] measures the consistency or coherence of explanations when handling similar instances, often calculated using the Lipschitz constant, which quantifies the sensitivity of AI explanations to input changes; lower value indicating higher stability [43], [344].

Specific Metrics: These sorts of metrics target specialized evaluations, such as assessing counterfactual explanations and concept learning models for their interpretability and adherence to human reasoning patterns. They could be classified as follows.

Counterfactual Validity: Counterfactual validity (CV) [365] ensures that the counterfactual explanation reflects a valid change in the prediction of the model and measures whether

a counterfactual explanation corresponds to a change in the prediction of the classifier. If the classifier predicts an image as normal, the counterfactual should be classified as abnormal [366], [367].

Frechet Inception Distance: Frechet inception distance (FID) [368] evaluates how visually similar or different the counterfactual explanation is compared to the original input. FID quantifies the visual quality of counterfactual explanations by calculating the feature distance between the original input image and the counterfactual image [369]–[372].

Foreign Object Preservation: Foreign object preservation (FOP) [368] ensures that the counterfactual explanation maintains the relevant details of the original input. FOP checks whether the counterfactual explanation retains individual patient information [67], [373].

Instance/Importance Metric: This will be divided into IM1 and IM2 metrics [374]. These metrics evaluate the consistency and quality of counterfactual explanations by analyzing reconstruction errors and similarities. While IM1 calculates the reconstruction error ratio between counterfactual instances, IM2 compares similarities among reconstructed counterfactual instances [366], [375]–[378].

Statistical Significance Test for Concepts: A statistical significance test for concepts helps avoid false results by ensuring that the concepts used are important and relevant to the class prediction. It checks the stability and importance of concept activation vectors in concept-based models. A statistical test, like a two-sided t-test, is used to see if the concept activation vectors are strongly related to class predictions [278], [379]–[384].

C. FAIR AI

Selecting appropriate fairness metrics is essential for evaluating equity in AI-enabled medical imaging models, as these metrics help identify and quantify biases, preventing AI systems from disproportionately disadvantaging specific groups. Researchers have developed a variety of fairness metrics to gain valuable insights into AI model performance, including demographic parity, equal opportunity, and predictive quality disparity [385], as illustrated in Figure 3). This section provides an overview of AI fairness metrics, with particular emphasis on their application in AI-powered medical imaging.

Equalized Odds: Equalized odds (EqOdd) [310] is a fairness metric that requires both the True Positive Rate (TPR) and the False Positive Rate (FPR) to be equal across different sensitive and/or protected groups (e.g., sex, age, race). This means that the model should make correct predictions (True Positives) and incorrect predictions (False Positives) at the same rate for each group, ensuring no group is disadvantaged in either aspect [298], [386]–[388].

Equal Opportunity: Equal opportunity (EO) [310] is a simpler version of Equalized odds. It focuses on equalizing the True Positive Rate (TPR) across different sensitive and/or protected groups. However, it does not require the rate of

incorrect positive predictions (False Positive Rate or FPR) to be the same across groups [306], [389]–[392].

Demographic Disparity: Demographic disparity (DP) [393] is a fairness metric that quantifies the percentage difference in positive outcomes across different demographic groups, measuring the diversity of positive outcomes for each sensitive and/or protected group [306], [394]–[396].

Predictive Quality Disparity: Predictive quality disparity (PQD) [385] measures the disparity in prediction quality among subgroups by calculating the ratio between the lowest and highest accuracy across those groups [306], [307], [385], [397].

Skewed Error Ratio: Skewed error ratio (SER) [398] evaluates fairness by measuring the skew or imbalance in error rates, such as false positives and false negatives. These are measured across specific groups by calculating the ratio of the highest to lowest error rate among those groups, with higher values indicating more significant bias and values closer to one reflecting lower bias [29], [60], [399]–[401].

Binary Cross Entropy: Binary cross entropy (BCE) is a loss function used to measure how well the predicted AI model's outcomes in a binary fashion align with the actual binary outcomes [66], [298], [402].

Expected Calibration Error: Expected calibration error (ECE) [403], [404] measures how well predicted probabilities align with true outcomes, indicating group sufficiency. While lower ECE confirms consistent threshold performance across different groups, higher ECE values may suggest a need for different optimal thresholds [66], [298], [312], [405].

Pairwise Fairness Difference: Pairwise fairness difference (PFD) [406] measures fairness by comparing the differences in AI model's outcomes between pairs of subgroups, where a large PFD indicates significant disparities and a lack of fairness in the AI model's predictions [407]–[410].

Equity-Scaled Dice Coefficient: Equity-scaled dice coefficient (ES-Dice) adjusts the Dice coefficient to account for performance disparities across subgroups, providing a straightforward evaluation and more interpretable measure of fairness by comparing overall and subgroup-specific Dice coefficients [68], [411], [412].

V. CASE STUDY: HIP AND KNEE BONY ANATOMY SEGMENTATION USING REF-AI

AI-powered segmentation of hip and knee bony anatomy is essential for pre-surgical planning, prosthesis design, and the evaluation of musculoskeletal disorders. However, challenges such as variations in patient anatomy, differences in imaging modalities, and demographic diversity raise important questions about the need for responsible and equitable AI solutions. As AI rapidly grows in healthcare, developing sophisticated AI systems that can address these challenges and improve equitable AI utilization for all patient groups becomes increasingly important. Furthermore, without adequate explainability, the adoption of AI in healthcare may be slowed, limiting its potential to improve outcomes across

diverse populations. That being said, the explainability of AI models is important to build trust and transparency in clinical settings, enabling healthcare providers to understand and validate AI-driven insights with a high level of confidence [413].

This case study explores how REF-AI principles can address these challenges by integrating fairness, transparency, and accountability into AI-driven image segmentation techniques. By leveraging insights and methodologies from our recent work in the literature [29], we aim to demonstrate how the application of REF-AI principles improves technical performance and embeds ethical considerations into the clinical use of AI.

A. METHODOLOGY

The pipeline for REF-AI in bony anatomy segmentation includes the following:

1) Data Governance

Data was sourced from the Osteoarthritis Initiative (OAI) dataset [414], a diverse imaging repository that includes varying demographics, imaging machines, and clinical sites, ensuring equitable representation of medical images. The OAI dataset is a longitudinal study that provides imaging such as plain knee radiographs, MRIs, and associated clinical outcomes. The OAI dataset's design includes standardized imaging protocols across multiple clinical sites, ensuring consistency and high-quality data collection. Furthermore, demographic diversity was carefully accounted for, with participants representing various ages, sexes, and ethnic backgrounds, allowing for equitable AI training and evaluation. OAI's data governance adhered to established privacy frameworks, including HIPPA, ensuring patient confidentiality. Comprehensive metadata documentation supported data traceability, enabling reproducibility and regulatory compliance. These qualitative and quantitative measures ensure that the dataset meets the ethical and technical requirements to develop fair and reliable segmentation models.

2) Algorithm Design

To develop advanced AI-powered segmentation models, we utilized a U-Net architecture with a pretrained ResNet18 backbone, leveraging transfer learning to improve feature extraction. To address class imbalances, we applied data balancing techniques such as undersampling of overrepresented classes. Stratified sampling was used to reflect the demographic diversity of the population in the training and testing datasets, particularly in terms of sex and race. Furthermore, we trained specialized AI models for specific protected attributes, enabling us to evaluate group-specific [29], [415] performance and implement targeted bias mitigation strategies. These strategies were designed to make the AI-powered segmentation models equitable and reliable, providing accurate results across diverse patient populations.

TABLE 2: The summary of REF-AI evaluation metrics.

Category	Evaluation Metric	Reference(s)
Explainable AI	Human Involvement	(Doshi et al., 2017) [322], (Muddamsetty et al., 2021) [323], (Hossain et al., 2023) [43]
	Pointing Game Metrics	(Zhang et al., 2018) [330], (Miró-Nicolau et al., 2022) [214], (Hossain et al., 2023) [43]
	Attribution Localisation Metrics	(Kohlbrener et al., 2020) [331], (Papanastasiou et al., 2023) [333], (Rio et al., 2024) [332]
	Top-k Intersection	(Ghorbani et al., 2019) [334], (Hossain et al., 2023) [43]
	Concept Influence Score	(Theiner et al., 2022) [335], (Hossain et al., 2023) [43]
	Relevance Mass Accuracy	(Arras et al., 2022) [336], (Srinivasan et al., 2022) [338], (Springenberg et al., 2023) [337], (Hossain et al., 2023) [43]
	Relevance Rank Accuracy	(Arras et al., 2022) [336], (Srinivasan et al., 2022) [338], (Wickstrom et al., 2023) [339], (Hossain et al., 2023) [43]
	Faithfulness Correlation	(Bhatt et al., 2020) [340], (Arras et al., 2022) [336], (Srinivasan et al., 2022) [338], (Komorowski et al., 2023) [341], (Wickstrom et al., 2023) [339], (Springenberg et al., 2023) [337], (Hossain et al., 2023) [43]
	Deletion and Insertion Metrics	(Samek et al., 2016) [342], (Saleem et al., 2021) [59], (Hu et al., 2022) [343], (Hossain et al., 2023) [43], (Nielsen et al., 2023) [344]
	Remove And Retrain (ROAR)	(Hooker et al., 2019) [345], (Hu et al., 2022) [343], (Nielsen et al., 2023) [344], (Sadafi et al., 2023) [346]
	Remove and Debias (ROAD)	(Rong et al., 2022) [347], (Hossain et al., 2023) [43]
	Area Over Perturbation Curve (AOPC)	(Samek et al., 2016) [342], (Kalipolitis et al., 2023) [349], (Gallo et al., 2023) [350], (Lamprou et al., 2024) [348]
	Local Lipschitz Estimate	(Alvarez et al., 2018) [351], (Doumard et al., 2023) [352]
	Region Perturbation	(Samek et al., 2016) [342], (Nam et al., 2020) [356], (Sun et al., 2023) [353], (Binder et al., 2023) [354], (Kadir et al., 2023) [355]
	Pixel-Flipping	(Bach et al., 2015) [241], (Pitroda et al., 2021) [358], (Farrag et al., 2023) [255], (Gnanavel et al., 2024) [357]
	Model Parameter Randomization	(Adebayo et al., 2018) [359], (Saleem et al., 2021) [59], (Gunasekar et al., 2022) [230], (Hossain et al., 2023) [43]
	Fidelity	(Guidotti et al., 2019) [360], (Velumuran et al., 2021) [361] (Huang et al., 2021) [363], (Nielsen et al., 2023) [344], (Buliga et al., 2024) [362]
	Stability	(Alvarez et al., 2018) [364], (Hossain et al., 2023) [43], (Nielsen et al., 2023) [344]
	Counterfactual Validity (CV)	(Mothilal et al., 2020) [365], (Verma et al., 2020) [366], (Verma et al., 2024) [367]
	Frechet Inception Distance (FID)	(Ghandeharioun et al., 2021) [372], (Singla et al., 2023) [368], (Lamiabie et al., 2023) [369], (koetzier et al., 2024) [370], (Xu et al., 2024) [371]
Fair AI	Foreign Object Preservation (FOP)	(Singla et al., 2023) [368], (Patricio et al., 2022) [373], (Patricio et al., 2023) [67]
	Instance/Importance Metric	(Mahajan et al., 2019) [375], (Verma et al., 2020) [366], (Van et al., 2021) [374], (Khorram et al., 2022) [378], (Labaen et al., 2023) [377], (Taylor et al., 2024) [376]
	Statistical Significance Test for Concepts	(Kim et al., 2018) [278], (Ghorbani et al., 2019) [379], (Yeh et al., 2020) [380], (Foscarin et al., 2022) [382], (Ma et al., 2024) [381], (Kowal et al., 2024) [383], (Chanda et al., 2024) [384]
	Equalized Odds (EqOdd)	(Hardt et al., 2016) [310], (Goel et al., 2018) [387], (Awasthi et al., 2020) [386], (Jung et al., 2021) [388] (Zong et al., 2022) [298]
	Equal Opportunity (EO)	(Hardt et al., 2016) [310], (Beutel et al., 2019) [391], (Narasimhan et al., 2020) [390], (Saha et al., 2020) [389], (Huang et al., 2022) [392], (Yang et al., 2023) [306]
	Demographic Disparity (DP)	(Dwork et al., 2012) [393], (Castelnovo et al., 2022) [394], (Yang et al., 2023) [306], (Ghosh et al., 2022) [396], (Cohausz et al., 2024) [395]
	Predictive Quality Disparity (PQD)	(Du et al., 2020) [385], (Du et al., 2022) [397], (Chiu et al., 2023) [307], (Yang et al., 2023) [306]
	Skewed Error Ratio (SER)	(Wang et al., 2020) [398], (Wang et al., 2020) [399], (Puyol-Antón et al., 2021) [60], (Siddiqui et al., 2024) [29], (Ohki et al., 2024) [400], (Atzori et al., 2024) [401]
	Binary Cross Entropy (BCE)	(Ruby et al., 2020) [402], (Zong et al., 2022) [298], (Zhang et al., 2022) [66]
	Expected Calibration Error (ECE)	(Guo et al., 2017) [403], (Pleiss et al., 2017) [312], (Nixon et al., 2019) [404], (Zong et al., 2022) [298], (Zhang et al., 2022) [66], (Brahmbhatt et al., 2023) [405]
	Pairwise Fairness Difference (PFD)	(Narasimhan et al., 2020) [406], (Hazirbas et al., 2021) [408], (Lin et al., 2023) [407], (Dodevska et al., 2023) [410], (Ferrara et al., 2024) [409]
	Equity-Scaled Dice Coefficient (ES-Dice)	(Tian et al., 2023) [411], (Masroor et al., 2024) [412], (Tian et al., 2025) [68]

3) AI Explainability using GradCAM

Grad-CAM [35], [36] was utilized to provide visual explanations of the segmentation process, enabling clinicians to understand the AI model. Grad-CAM highlighted regions of interest in the imaging data that most influenced the model's predictions, offering clear transparency into the decision-making process. Grad-CAM visualizations, shown in Figures 4 and 5 demonstrate the potential to align AI outputs with clinician expectations and build trust in the system.

4) Fairness Evaluation

The fairness of the various strategies was evaluated using the skewed error ratio (SER) [398]. This metric quantifies bias towards protected attributes by measuring the disparities in the model's prediction errors. SER is calculated by dividing the highest error rate to the lowest error rate among the protected or sensitive groups. A higher SER value indicates

greater bias, while values closer to one reflect minimal bias and greater fairness in model performance.

B. RESULTS

The application of the REF-AI framework to hip and knee bony anatomy segmentation yielded promising results. By integrating diverse demographic data from the OAI dataset [414], the study achieved broad representation and equitable model performance. The advanced segmentation models demonstrated robust performance in identifying bony structures across varying patient demographics. Bias mitigation strategies minimized disparities across protected demographic groups, as indicated by the SER values in Tables 3 - 6. Furthermore, Grad-CAM visualizations, shown in Figures 4 and 5 successfully highlighted key anatomical landmarks. These visualizations aligned the AI model outputs with clinician expectations, fostering trust in the system.

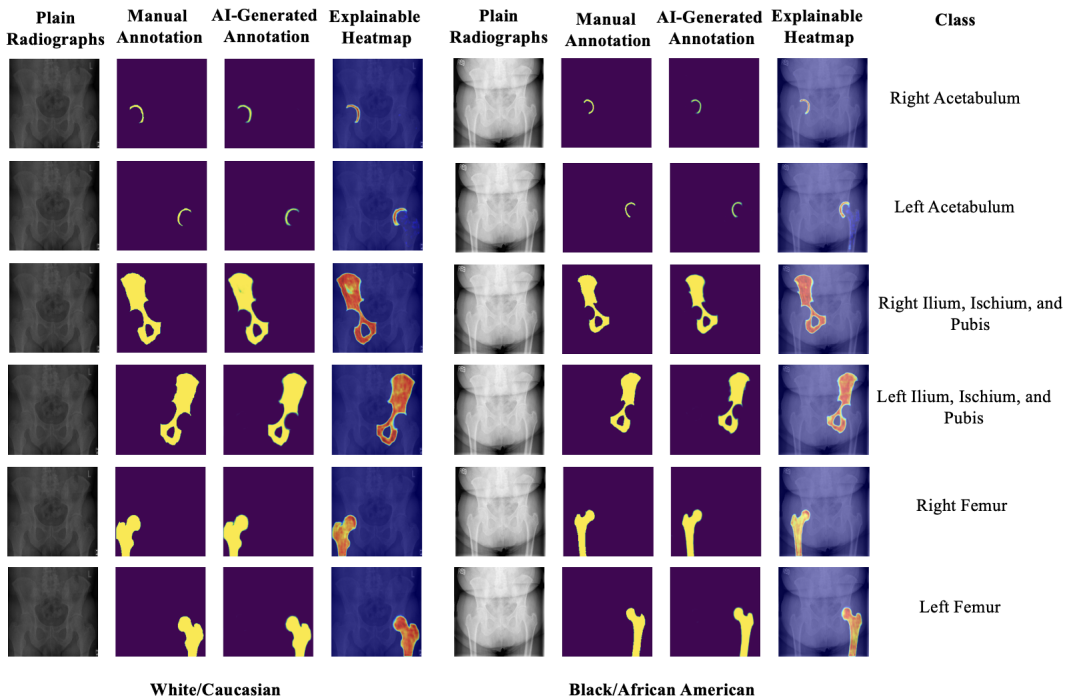


FIGURE 4: Generated Grad-CAM heatmaps for hip bony anatomy segmentation across the protected attribute race.

Analysis of Tables 3 - 6 reveals key trade-offs between segmentation accuracy (measured by IoU) and fairness (evaluated using SER). Specifically, incorporating fairness into the segmentation algorithm often reduces performance for either one or both protected attribute groups. These trade-offs can be seen across all the results. For example, in Table 5 the Baseline model achieves higher IoU scores but demonstrates a higher disparity in fairness, with an SER of 1.112 across racial groups. On the other hand, while improving fairness (SER=1.050), the group-specific strategy [29] results in a reduction in IoU for the protected subgroups, demonstrating that achieving equitable performance often necessitates compromising overall segmentation precision.

Furthermore, the results demonstrate that no single bias mitigation strategy is universally optimal across all protected groups. For example, for hip segmentation, the Balanced model performs best for sex, whereas the Group-specific model is optimal for race. In contrast, for knee segmentation, the Stratified model yields the best performance for both sex and race.

C. CASE STUDY HIGHLIGHT

This case study demonstrates the significance of integrating REF-AI principles in medical imaging. The results emphasize the value of comprehensive data governance, with OAI's standardized imaging protocols and privacy frameworks ensuring high-quality, ethically compliant datasets. Targeted bias mitigation strategies significantly addressed disparities related to demographic diversity, including variations in sex and race attributes. These efforts enhanced fairness and

Model	Male IoU	Female IoU	SER	SD
Baseline	0.870	0.868	1.014	0.000
Balanced	0.851	0.852	1.007	0.000
Stratified	0.870	0.862	1.054	0.000
Group-specific	0.853	0.847	1.044	0.002

TABLE 3: IoU and fairness scores for hip segmentation across the protected attribute sex.

Model	Male IoU	Female IoU	SER	SD
Baseline	0.924	0.925	1.021	0.001
Balanced	0.899	0.909	1.107	0.005
Stratified	0.921	0.920	1.015	0.001
Group-specific	0.916	0.920	1.050	0.002

TABLE 4: IoU and fairness scores for knee segmentation across the protected attribute sex.

contributed to more inclusive and representative AI model performance. Moreover, AI explainability tools such as Grad-CAM provided critical insights into the model's decision-making process. These scientific visualizations allow researchers and clinicians to better understand how the AI model arrived at its predictions, enhancing transparency and alignment with clinical reasoning. This transparent approach builds trust among healthcare professionals and bridges the gap between AI-driven insights and real-world clinical applications.

Overall, this case study illustrates how the thoughtful application of REF-AI principles can enhance the technical robustness and ethical integrity of AI-powered medical imaging solutions, ultimately improving patient and clinical outcomes and equity in healthcare.

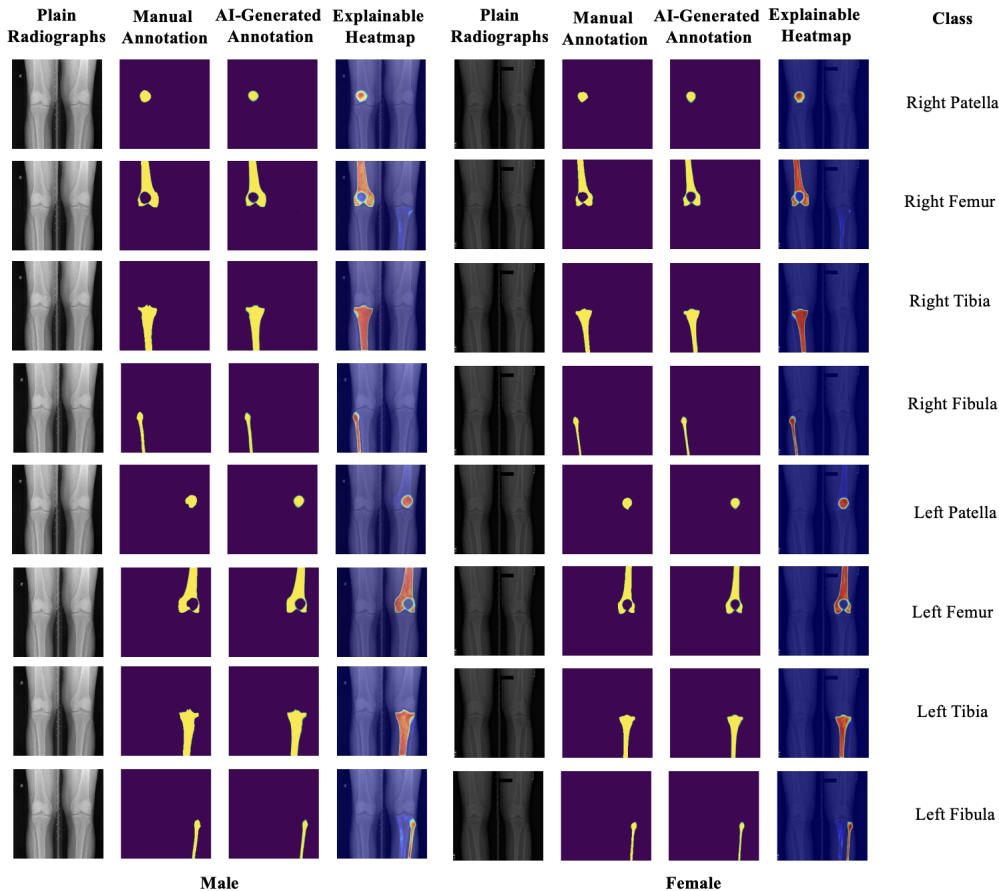


FIGURE 5: Generated Grad-CAM heatmaps for knee bony anatomy segmentation across the protected attribute sex.

Model	White/Caucasian IoU	Black/AA ¹ IoU	SER	SD
Baseline	0.875	0.861	1.112	0.007
Balanced	0.856	0.847	1.062	0.004
Stratified	0.864	0.849	1.102	0.007
Group-specific	0.853	0.846	1.050	0.003

TABLE 5: IoU and fairness scores for hip segmentation across the protected attribute race.

Model	White/Caucasian IoU	Black/AA ² IoU	SER	SD
Baseline	0.924	0.925	1.015	0.000
Balanced	0.902	0.921	1.229	0.009
Stratified	0.916	0.916	1.000	0.000
Group-specific	0.919	0.915	1.055	0.002

TABLE 6: IoU and fairness scores for knee segmentation across the protected attribute race.

VI. DISCUSSION AND OUTLOOK

This paper explores the integration of responsibility, explainability, and fairness in AI, particularly within medical image analysis. A few years ago, the primary focus in AI development was mainly on improving AI model accuracy. However, as AI continues to play an increasing role in healthcare, it is clear that the conversation should be expanded. Today, accu-

racy for AI algorithms alone is no longer sufficient, meaning that the success of AI models in healthcare hinges on their performance combined with ethical considerations, especially in responsibility, explainability, and fairness. These factors are now essential for deploying AI technologies in a way that is effective and socially acceptable.

The AI fairness strategies, including pre-processing, in-processing, and post-processing, each offer distinct advantages, but their real-world application remains challenging. Pre-processing methods, such as group rebalancing and data augmentation, help mitigate biases in datasets before the AI model is trained, but they must be carefully calibrated to avoid overfitting and enhance the AI model's generalizability. In-processing strategies, such as adversarial training and fairness constraints, adjust the learning process to build equitable predictions. However, they often increase computational costs and, in some cases, may impact AI model accuracy. Post-processing techniques, such as equalized odds and reject option classification, allow for the calibration of model outputs after training to fine-tune fairness across various protected and/or sensitive subgroups. However, these methods alone cannot fully fix potential biases in the data and AI, thus a more complete and systematic approach to fairness is needed. Fairness in AI will not be obtained in isolation. The

¹African American²African American

challenges of mitigating biases related to sex, race, ethnicity, age, economic situation, and level of healthcare education persist in medical imaging, and efforts to enhance fairness must be balanced with the need for AI model performance. This balance is particularly challenging in healthcare, where the stakes are high and any compromise in AI model accuracy can have profound consequences. Furthermore, the need for fairness must be considered in the context of broader societal values, which can vary significantly across different cultures, states, and community settings. This underscores that addressing fairness in AI is not merely a technical task, it is however an ethical imperative that requires continuous collaboration and dialogue among diverse stakeholders.

Explainability also plays a significant role in the responsible use of AI systems. As AI becomes more popular in healthcare systems, it is essential that the AI models are transparent and their decision-making processes are easy to understand. This will help to build trust among all end-users. The evaluation of AI explainability must consider both subjective and objective metrics. Subjective metrics ensure that AI explanations are aligned with human understanding, while objective methods, such as attribution-based, offer a more quantitative understanding of how models make decisions. However, these metrics must be applied carefully to avoid oversimplifying or overcomplicating complex AI models.

Responsibility in AI extends beyond fairness and explainability; it requires ethical principles that guide the entire AI lifecycle. Developing standardized metrics for responsible AI remains a significant challenge. Current frameworks often focus on qualitative evaluations, such as stakeholder engagement and ethical compliance, but there is a need for more solid and universally applicable standards. One promising direction is the integration of fairness, accountability, and transparency frameworks, which could help establish consistent guidelines for responsible AI across different healthcare domains. In medical imaging, where the impact of AI on human lives is profound, creating universally accepted standards that balance fairness, accountability, and transparency will be significant for guiding ethical decision-making.

Building truly responsible, explainable, and fair AI (REF-AI) in medical imaging informatics will require a collaborative and multidisciplinary framework that engages all parties and users. This includes technical experts, developers, AI programmers, patients, caregivers, healthcare providers, clinicians, surgeons, physicians, nurses, policy-makers, ethicists, and other relevant groups. Such collaboration is essential to implementing AI systems that reflect diverse needs, values, and ethical considerations.

In conclusion, the intersection of responsibility, explainability, and fairness is essential for developing AI systems that are ethical, transparent, and trustworthy, particularly in high-risk domains, such as medical image analysis. While the advancement of AI fairness offers promising solutions to detect and mitigate biases, their real-world application requires careful consideration of trade-offs between fairness and AI model performance. The role of explainability

is similarly critical, helping to build AI systems that are transparent and aligned with human understanding. Despite significant progress, challenges remain in harmonizing these three pillars of AI responsibility, explainability, and fairness in complex domains in healthcare. Further research should prioritize refining these strategies, addressing the ethical dilemmas associated with their application, and establishing a standardized framework. This framework would be instrumental in developing comprehensive metrics for the design, development, evaluation, and implementation of REF-AI, ensuring that these systems are both effective and ethically sound.

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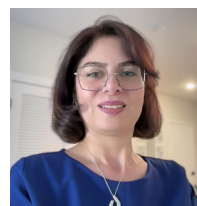
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